

Master Project State File

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Underground CBRN-Hardened Blast-Resistant Protective Structure + Sentry Post — Pune, Maharashtra

Portable memory · Source of truth · Revision record · Reconstruction specification

Document status: MASTER STATE, issue 1 **Compiled:** 2 September 2026 **Compiled from:** Phase 1 Design Report Rev D (126 KB, 2000 lines), ten Rev F architectural DXF files, nineteen STAAD.Pro screen captures, and the full Phase 2 structural design work. **Covers:** Phase 1 (architectural + basis of design, complete) and Phase 2 (structural design + drawings, substantially complete). **Next phase:** Phase 3 — non-linear SDOF verification, site investigation close-out, sentry post drawing S-07 equivalent.

⚠ EVIDENCE-CLASS LEGEND — APPLIED THROUGHOUT THIS DOCUMENT

Tag	Meaning	How a future Claude must treat it
[CONFIRMED]	Read directly from an uploaded file, or computed in this project from confirmed inputs	Use as fact
[RECONSTRUCTED]	Derived by back-calculation from partial evidence (e.g. STAAD statics-check output)	Use, but state the derivation when relied on
[ASSUMED]	Engineering assumption, no test or client confirmation	Must be re-confirmed before construction
[UNRESOLVED]	Two or more conflicting values, insufficient evidence to choose	DO NOT GUESS. Ask the user.
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PART A — CURRENT AUTHORITATIVE PROJECT STATE

Everything in Part A is the **LATEST VALID DESIGN**. If any uploaded file disagrees with Part A, stop and raise the conflict before proceeding.

A.1 Project identity

Field	Value	Class
Project title	Underground CBRN-hardened, blast-resistant protective structure with associated sentry post	[CONFIRMED]
Location	Pune, Maharashtra, India	[CONFIRMED]
Project type	Buried reinforced-concrete protective structure, military-operational, for actual construction	[CONFIRMED]
Academic context	B.E. Civil Engineering, final semester, three-presentation capstone	[CONFIRMED]
Objective	Protect 9 occupants for 96 h against a nuclear air-blast design basis threat with CBRN, EMP and fallout hardening	[CONFIRMED]
Architectural revision	Rev F (the ten uploaded DXF files)	[CONFIRMED]
Design report revision	Rev D (older than the drawings — this is the single largest reconciliation issue)	[CONFIRMED]
Structural revision	Phase 2 Rev A, incorporating Modification M1	[CONFIRMED]
Presentation status	P1 (preliminary/architectural) delivered. P2 (AutoCAD + STAAD + manual calculations) is the active deliverable. P3 not yet scoped	[CONFIRMED]
Software in use	STAAD.Pro (Bentley), AutoCAD/DXF, plus a purpose-built dependency-free Python DXF toolchain (see Part F)	[CONFIRMED]
Presentation window	~90 minutes, mixed academic + military audience	[CONFIRMED]

A.2 Overall configuration

Three separate structures:

1. **Main underground box** — buried RC shelter, 8 bays, 22.0 × 6.2 m external, roof 2.0 m below grade.
2. **Entry headhouse + covered approach stairwell** — above-ground bermed RC, sitting on the shelter roof and on adjacent fill.
3. **Sentry post** — a separate two-storey RC framed building, **≥ 10 m clear** of the shelter excavation, on its own footings on in-situ rock. **Not blast designed** (declared expendable).

Access / egress: - **Primary:** grade → covered entry stairwell (12R @ 166.667/300) → platform (−2.000) → inner security door → headhouse → stair void → main dog-leg stair (24R @ 170.833/280, 3 flights × 8) → arrival landing (−6.100) → **Blast Door 1** (1200 × 2100, 7 bar) into Bay 6. - **Secondary:** **Blast Door 2** (1200 × 2100) from the stair shaft into Bay 8 (generator/grey zone). - **Emergency:** two 1400 mm dia vertical escape shafts, **ESC 1** and **ESC 2**, with 250 mm RC collars, in Bays 1 and 8.

PROTECTIVE BOUNDARY = Blast Doors 1 and 2 at level (−)6.100, plus walls W6/W7, the perimeter walls, the mat and the pressure slab. Everything above the blast doors (stair shaft, headhouse, entry stairwell) is OUTSIDE the boundary and the entry stairwell is DECLARED EXPENDABLE.

Gas-tight envelope = Bays 1 to 6 only — 67.8 m² floor, 217.0 m³ volume. Bays 7 (stair shaft) and 8 (generator, “grey zone”) are outside it.

A.3 Bay schedule — CURRENT (after Modification M1)

Bay	X range (mm)	Clear width	Use
1	600 - 3500	2900	Emergency stores, 1000 L potable tank, ESC 1
2	3610 - 5410	1800	Lavatory (1800×2000) + medical (1800×3000)
3	5520 - 9020	3500	Ops room & hazard plotting; EMP Zone 2 enclosure
4	9130 - 10930	1800	Berthing, 3 × 3-tier bunks, 9 berths
5	11040 - 12600	1560	CBRN plant: 2 × 250 m³/h filters, CO₂/O₂, dehumidifier, sump
6	12800 - 14800	2000	Decon airlock, 3 stages (2000×2000, 2000×1500, 2000×1500)
7	15200 - 18000	2800	Stair shaft (shifted +200 by M1)
8	18400 - 21400	3000	Generator 15 kVA (grey zone), ESC 2 at X 19900

Internal wall positions — CURRENT:

Mark	X range (mm)	Thickness	Notes
W8 partitions ×4	3500-3610, 5410-5520, 9020-9130, 10930-11040	110	Each with a 900 door gap at Y 2500-3400
W5	12600 - 12800	200	Bay 5/6, fire + gas-tight only, no pressure differential
W6	14800 - 15200	400	Bay 6/7 — MOD M1, was 200. Protective boundary. Blast Door 1 opening Y 600-1800
W7	18000 - 18400	400	Bay 7/8 — MOD M1, was 200. Protective boundary. Blast Door 2 opening Y 600-1800

A.4 Geometry — CURRENT AUTHORITATIVE VALUES

A.4.1 Coordinate system (used by every DXF and every calculation in this project)

ORIGIN = south-west EXTERNAL corner of the underground box, at ground level
 X = east (along the length, 0 to 22000)
 Y (plan) = north (across the width, 0 to 6200)
 LEVELS = metres relative to finished site grade 0.000, negative downwards
 UNITS = millimetres in all DXF geometry; metres in all level annotation

A.4.2 Main box

Item	CURRENT value	Was	Class
External length	22 000 mm	21 600 (pre-M1)	[CONFIRMED]
External width	6 200 mm	unchanged	[CONFIRMED]
Internal length	20 800 mm	20 400	[CONFIRMED]
Internal width (clear span of the roof)	5 000 mm	unchanged — do not widen	[CONFIRMED]
Perimeter wall thickness	600 mm	unchanged	[CONFIRMED]
Roof (pressure) slab	900 mm	unchanged	[CONFIRMED]
Mat foundation	600 mm	unchanged	[CONFIRMED]
Blinding / PCC	100 mm M15	unchanged	[CONFIRMED]
Internal clear height	3 200 mm	unchanged	[CONFIRMED]
Engineered cover over roof	2 000 mm layered	reduced from 4 000 in an earlier revision	[CONFIRMED]

A.4.3 Level schedule — CURRENT

Level	Value	Description
Grade	0.000	Finished site level, crowned, falls 1:50 away
Top of pressure slab	(-)2.000	= headhouse floor level
Roof soffit	(-)2.900	900 slab
Internal floor / top of mat	(-)6.100	3200 clear height
Underside of mat	(-)6.700	600 mat
Formation / underside of PCC	(-)6.800	100 blinding
Design groundwater table	(-)2.000	monsoon [ASSUMED — requires monsoon monitoring]
Rockhead	(-)1.500 to (-)2.000	[ASSUMED]
Stair landing L1	(-)4.7333	8 risers up from floor
Stair landing L2	(-)3.3667	16 risers up from floor
Headhouse roof soffit	+0.400	2400 clear internally
Headhouse roof top	+0.900	no earth cover; berm graded to this level
Entry stairwell roof at head	+2.450	soffit +2.200
Sentry post GF FFL	+0.450	= base of the STAAD model
Sentry post first floor	+3.650	storey height 3200
Sentry post roof	+6.700	storey height 3050
Sentry post parapet top	+7.000	
Sump pit invert	(-)7.600	pit base slab (-)8.000

A.4.4 Stair shaft, void and pad — CURRENT (post-M1)

Stair shaft (Bay 7) clear X 15200 – 18000 Y 600 – 5600 = 2800 × 5000
 Void in the pressure slab X 15200 – 18000 Y 600 – 3760 = 2800 × 3160
 Cantilever pad (remaining slab) X 15200 – 18000 Y 3760 – 5600 = 2800 × 1840
 Flight A (flights 1 and 3, stacked) X 15300 – 16500 (1200 wide)
 Well X 16500 – 16700 (200)
 Flight B (flight 2) X 16700 – 17900 (1200 wide)
 Landing L1 (-)4.7333 Y 3760 – 4960, full 2800 width
 Landing L2 (-)3.3667 Y 600 – 1800, full 2800 width (stacked over the arrival landing)
 Arrival landing (-)6.100 = the mat surface, Y 600 – 1800
 Store under landing L1 Y 4960 – 5600
 Main stair: 24R @ 170.8333, tread 280, 3 flights × 8R, total rise 4100, headroom 2533

A.4.5 Escape shafts

	ESC 1	ESC 2
Centre	(2050, 2050)	(19900, 2050) — shifted +400 by M1
Clear opening	1400 mm dia	1400 mm dia
Collar	250 mm RC, OD 1900	250 mm RC, OD 1900
Clearance, opening edge to wall face	750 mm all round	800 / 800 mm

THE 750 mm CLEARANCE RULE. Opening edge to structural wall face ≥ 750 mm = 250 collar + ~400 trimmer band + 100 tolerance. **A bay must therefore be at least 1400 + 2 × 750 = 2900 mm wide to hold an escape shaft.** This rule drove the Rev C lengthening and it is why Modification M1 could not take 400 mm out of Bay 8.

A.4.6 Headhouse — CURRENT (shifted +200 by M1)

External X 13600 – 18400 Y 200 – 6000 = 4800 × 5800
Internal X 14000 – 18000 Y 600 – 5600 = 4000 × 5000
Walls 400 thk (HW1 south, HW2 north, HW3 west, HW4 east)
Roof 500 thk, soffit +0.400, top +0.900, NO earth cover
Floor = the top of the 900 pressure slab, (-)2.000
Clear internal height 2400
Inner security door 900 × 2100 in HW2 (north), at X 14450 – 15350. NOT blast rated.
Berm graded against all four walls to +0.900 at 1.5 : 1

Load path of each headhouse wall — [CONFIRMED, checked in Phase 2]:

Wall	Runs	Length	Sits over
HW1 south	in X	4000	Box south perimeter wall — direct ✓
HW2 north	in X	4000	Box north perimeter wall — direct ✓
HW3 west	in Y	5000	Bay 6, mid-slab. NO wall below — line load on the pressure slab, checked at 26 % (two-way) / 41 % (one-way bound)
HW4 east	in Y	5000	Wall W7 (18000-18400). M1 made these align exactly ✓

A.4.7 Covered entry stairwell — CURRENT (Rev F, shifted +200 by M1)

External X 9250 – 16050 Y 5750 – 7750 = 6800 × 2000
Internal X 9500 – 15800 Y 6000 – 7500 = 6300 × 1500
Walls 250 RC (both sides, headwall, east wall)
Top landing X 9500 – 11000 at 0.000, 250 thk
Flight X 11000 – 14300, 12R @ 166.6667, going 300, 11 goings × 300 = 3300, waist 250
Platform X 14300 – 15800 at (-)2.000, 1500 × 1500, 250 thk
Roof 250 RC raking, soffit 2200 above the flight; over the platform it becomes the 500 headhouse roof
Entry door 1000 × 2100 at grade in the headwall (X 9250 – 9500), opens outward
300 mm channel + grating full 1500 width at X 8950 – 9250
Stepped RC raft 300 thk on compacted fill; MOVEMENT JOINT where it meets the headhouse
1.0 m³ external sump at the platform, own soakaway
Berm 1.5:1 against the walls to +0.900, toe at grade over 1350

A.4.8 Sentry post — CURRENT [all CONFIRMED from the Rev F DXFs]

External plan 4000 × 5000 Internal 3600 × 4600
Column grid A-B 3650 c/c (X) · 1-2 4650 c/c (Y)
Grid coordinates A: x = 175 B: x = 3825 | 1: z = 175 2: z = 4825
Columns C1 350 × 350, 4 No., both storeys
Beams B1 250 × 450, spanning A-B (3650 c/c), on grids 1 and 2
Beams B2 250 × 450, spanning 1-2 (4650 c/c), on grids A and B
Slab S1 150 thk two-way, 3650 × 4650 c/c, both floors
Plinth beam PB 250 × 400 at +0.450
Footings F1 1500 × 1500 × 600, 4 No., on in-situ basalt at (-)2.000
Ground storey infill 200 RC ballistic panels
First storey infill armoured vision panels, 1200 wide
Spiral stair external, 1000 R, 250 dia central pole
Door D1 900
Storey heights ground 3200 (+0.450 → +3.650), first 3050 (+3.650 → +6.700)
Siting >= 10 m clear of the shelter excavation

A.5 Materials — CURRENT

Item	Shelter / stairs / headhouse	Sentry post	Source
Concrete	M35	M30	IS 456 Table 5 (M35 = minimum for very severe exposure, Table 3)
w/c ratio	≤ 0.45	—	IS 456 Table 5
Cement content	≥ 340 kg/m³	—	IS 456 Table 5
Admixture	Integral crystalline waterproofing	—	[R]
Blinding	M15, 100 thk	—	
Burster slab (in the cover)	M30, 200 thk	—	
Reinforcement	Fe500D to IS 1786:2008	Fe500	
E _c = 5000/√f _{ck}	29 580 N/mm²	27 386 N/mm²	IS 456 Cl. 6.2.3.1
Poisson's ratio	0.20	0.20	[ASSUMED — standard]
Unit weight RC	25 kN/m³	25 kN/m³	IS 875 (Pt 1) Table 1
γ _m concrete / steel	1.5 / 1.15	same	IS 456 Cl. 36.4.2

Cover — IS 456 Cl. 26.4.2 / Table 16 [CONFIRMED]

Face	Cover
Cast against blinding / trimmed rock	75 mm
Formed earth face	50 mm
Internal faces	40 mm
Stair shaft faces (wet/dirty zone)	30 mm
Sentry post beams and slabs	30 mm
Sentry post columns	40 mm (Cl. 26.4.2.2 — never less than 40 nor the bar dia)
Footings cast against earth	50 mm (Cl. 26.4.2.1)
Max bar spacing, both curtains	150 mm — EMP requirement , stricter than IS 456 Cl. 26.3.3

The 5 mm cover reduction that IS 456 Table 16 permits for M35 and above is **deliberately not taken**.

Development and lap lengths — IS 456 Cl. 26.2.1 [CONFIRMED]

$L_d = \phi \sigma_s / (4 \tau_{bd})$, $\sigma_s = 0.87 f_y = 435 \text{ N/mm}^2$
 τ_{bd} (Cl. 26.2.1.1): M30 $\rightarrow 1.5$ M35 $\rightarrow 1.7$; $\times 1.6$ for deformed bars in tension
 $\div 0.8$ (i.e. $\times 1.25$) for bars in compression

Grade	$\tau_{bd} \times 1.6$	L_d tension	L_d compression
M35	2.72	40 ϕ	32 ϕ
M30	2.40	46 ϕ	37 ϕ

Bar	L_d M35	Lap M35	L_d M30	$L_{d,c}$ M30
T8	320	400	368	296
T10	400	500	460	370
T12	480	600	552	444
T16	640	800	736	592
T20	800	1000	920	740
T25	1000	1250	1150	925

Lap policy: IS 456 Cl. 26.2.5.1(c) gives lap = L_d or 30ϕ whichever greater = 40ϕ . Cl. 26.2.5.1 requires $\times 1.4$ if more than 50 % of bars are lapped at one section. **All laps staggered so ≤ 50 % are spliced at any section, and lap specified as 50ϕ — 25 % above requirement.**

Blast bond: IS 4991 Cl. 10.3.1.1 permits +25 % on bond ($\rightarrow 32\phi$). **Not taken.** All detailing uses the static 40ϕ .

A.6 Soil and foundation parameters — CURRENT

Parameter	Value	Class
Ground	Deccan basalt (trap), with red-bole / vesicular seams at flow contacts	[ASSUMED]
Rockhead	(–)1.500 to (–)2.000	[ASSUMED]
Presumptive SBC	3240 kPa (IS 1904:1986 Table 1, hard rock)	[ASSUMED]
Bulk unit weight γ	20 kN/m ³	[ASSUMED]
Saturated unit weight γ_{sat}	21 kN/m ³	[ASSUMED]
Submerged $\gamma' = 21 - 9.81$	11.19 kN/m ³	[CONFIRMED, derived]
At-rest coefficient $K_0 = 1 - \sin \phi$	0.50 ($\phi \approx 30^\circ$)	[ASSUMED]
Design GWT	(–)2.000 (monsoon)	[ASSUMED — the single most important number to confirm]
Modulus of subgrade reaction k_s	100 000 to 500 000 kN/m³	[ASSUMED] — RUN THE MAT MODEL AT BOTH BOUNDS
Settlement	Negligible on sound basalt (IS 12070)	[ASSUMED]
Soil-transmission factor K_a , saturated	1.0 (IS 4991 Cl. 7.2 + Table 3)	[CONFIRMED]
K_a , dry compacted berm fill	~0.5	[ASSUMED — deliberately not relied on , see C10]

The hazard is not the basalt — it is the flow contacts. A single red-bole seam under the mat produces the differential-support case that SIZES the mat. Over-excavate any red-bole or vesicular seam at founding level and replace with M15 lean concrete.

A.7 Loading — CURRENT, complete register

A.7.1 Blast

DBT nuclear air-blast, $p_{so} = 344.7 \text{ kPa}$ (50 psi) [C]
Positive phase t_d 0.13 to 1.33 s [C]
Reflected pressure p_r 1366 kPa [CONFIRMED]
Dynamic pressure q 282 kPa [CONFIRMED]
Ductility ratio μ 5 (moderate, repairable damage) IS 4991 Cl 10.3.3
 $DLF = \mu/(\mu - 0.5)$ $5/4.5 = 1.111$
validation: $\mu = 1 \rightarrow DLF = 2.00 =$ the textbook step-load factor ✓
DESIGN BLAST PRESSURE $344.7 \times 1.111 = 383 \text{ kPa}$
applied to the roof AND the walls ($K_a = 1.0$)
Roof natural period T 13.4 ms $\rightarrow t_d/T = 10$ to 100 $\rightarrow$ QUASI-STATIC
Dynamic material strengths (BLAST CASE ONLY, IS 4991 Cl 10.3.1):
 $f_{ck,dyn} = 1.25 \times 35 = 43.75 \text{ N/mm}^2$ $f_{y,dyn} = 1.25 \times 500 = 625 \text{ N/mm}^2$
NO dynamic increase on shear – IS 4991 Cl 10.3.1.1

A.7.2 Gravity, soil and water

Load	Value	Applied to	Source
Self weight RC	25 kN/m ³	all	IS 875 (Pt 1) Table 1
Roof self, 0.900 × 25	22.50 kPa	roof	
Mat self, 0.600 × 25	15.00 kPa	mat	
Engineered cover, 2000 layered	40.65 kPa	roof	[CONFIRMED, see A.7.3]
SIDL services/finishes	2.0 kPa roof / 1.0 kPa mat		[R]
Live load, internal floor	5.0 kPa (plant/storage, not 2.0 residential)	floor	IS 875 (Pt 2)
Earth + water lateral gradient	$K_a \gamma' + \gamma_w = 0.50 \times 11.19 + 9.81 =$ 15.41 kPa/m	walls	[CONFIRMED, derived]
— at roof soffit (–)2.900	33.9 kPa	walls	
— at floor (–)6.100	83.2 kPa	walls	
Hydrostatic uplift on mat	$4.700 \times 9.81 =$ 46.11 kPa ; total 6289 kN over 136.4 m ²	mat	[CONFIRMED]
Construction surcharge	20 kPa vertical / 10 kPa lateral		[ASSUMED]
Staircase, smeared on the two shaft walls	2.947 kPa	W6/W7	[CONFIRMED — report LC5]

Of the 15.41 kPa/m lateral gradient, **9.81 is water and only 5.60 is soil**. Water is nearly two-thirds of the lateral load. This is why the groundwater level is the most important number the site investigation must confirm.

A.7.3 Engineered cover build-up — 40.65 kPa [CONFIRMED]

From the top	Thickness	Unit weight	Load kPa	Function
Topsoil / turf	300	18	5.40	Concealment, erosion, sheds rain
Granular filter	150	19	2.85	Stops fines clogging
RC burster slab M30, T12 @ 150 B/W	200	25	5.00	Breaks up a penetrating item
Crushed basalt rubble 25–75 mm	500	17	8.50	Scatters burster energy
Compacted engineered fill @ 95 % MDD	750	20	15.00	Radiation mass
Protection screed over the membrane	100	24	2.40	Protects waterproofing
TOTAL	2000		40.65	

Why 2.0 m: not blast (a buried roof takes full p_{so} regardless — IS 4991 Cl. 7.2) and not fallout (1.0 m already gives $PF \approx 2200$ against a requirement of ~ 1000). **The second metre is bought entirely for prompt neutron and gamma attenuation**, which needs mass and needs it above the slab. Reducing 4.0 m $\rightarrow$ 2.0 m saved 2 m of rock excavation, 2 m of shaft, one stair flight and 2 m of headroom.

A.7.4 Total roof load — COMB 103

Component	kPa
Blast, 344.7×1.111	383.00
Engineered cover	40.65
SIDL	2.00
Self weight	22.50
TOTAL w	448.15

Live load excluded — **IS 4991 Cl. 11.2:** “No live load shall be considered on roof at the time of blast.” Static ULS 101 on the roof = $1.5 \times (40.65 + 2.0 + 22.5 + 20) = 127.7 \text{ kPa} \rightarrow$ **blast governs 3.51 : 1.**

A.7.5 Headhouse loads

Element	Load	Basis
Roof (flush horizontal at berm crest)	396.5 kPa = 383 blast + 12.5 self + 1.0 SIDL	IS 4991 Cl. 7.2
Walls — CURRENT	383 kPa, acting EITHER FACE	Conflict C10 — raised from the report's 113 kPa drag
Walls — report basis (superseded)	113 kPa = C _d 'q = 0.4 × 282	IS 4991 Cl. 7.4

The walls sit behind ~2.5 m of berm and **earth transmits pressure**. K_a for dry compacted fill is an assumption we cannot verify, so the walls are designed for the full 383 kPa upper bound. **Cost of the upgrade: one link cage (T12 4-leg @ 250). Utilisation rises 17 % → 59 %.**

A.7.6 Entry stairwell loads — static only, NOT blast rated

Element	Load
Flight, waist 250	waist 7.150 + steps 2.083 + finishes 1.000 + LL 5.000 = 15.233 → w_u = 22.85 kPa
Side walls 250	K _o 0.50 × 20 × 2.9 + 0.5 × 10 surcharge = 34 kPa at base
Raking roof 250	self 6.25 + wp 2.0 + earth lap 5.4 + imposed 20 = 33.65 → w_u = 50.5 kPa
Top landing 250	18.4 + flight reaction 36.5 = 54.9 kPa

A.7.7 Sentry post loads — [CONFIRMED from the Rev F framing plan, all verified]

Item	Value	Verification
Slab dead, first floor	4.750 kPa	0.150 × 25 + 1.0 finish ✓
Slab dead, roof	5.250 kPa	3.75 + 1.5 screed/wp ✓
Imposed, first floor	3.000 kPa	IS 875 (Pt 2), observation post ✓
Imposed, roof	1.500 kPa	IS 875 (Pt 2), accessible ✓
Peak two-way intensity, DL floor	8.669 kN/m	4.75 × 1.825 ✓
Peak two-way intensity, DL roof	9.581 kN/m	5.25 × 1.825 ✓
Peak two-way intensity, LL floor	5.475 kN/m	3.00 × 1.825 ✓
Peak two-way intensity, LL roof	2.737 kN/m	1.50 × 1.825 ✓
Infill on first-floor beams	13.000 kN/m	0.200 × 2.600 × 25 ✓
Roof projection + parapet	4.162 kN/m	[C] as drawn, not independently derived
w _u floor = 1.5(4.75 + 3.00)	11.625 kPa	governs the slab
w _u roof = 1.5(5.25 + 1.50)	10.125 kPa	

Load path to the beams — IS 456 Cl. 24.5, 45° yield lines:

Peak intensity on any beam = w × short span/2 = w × 3.650/2 = w × 1.825 m
SHORT beams B1 (3650 long) → TRIANGLE
LONG beams B2 (4650 long) → TRAPEZOID, w_{eq} = w_{peak} × [1 - 1/(3r²)], r = 1.274 → 0.7946

A.7.8 Seismic

Underground box — IS 1893 (Part 1):2016

Z = 0.16 (Zone III) I = 1.5 R = 4.0 Sa/g = 2.5
Ah = (Z/2)(I/R)(Sa/g) = 0.08 × 0.375 × 2.5 = 0.075
W = 14 002 kN → Vb = 1050 kN
Per long wall 525 kN → τ = 525e3/(600 × 0.8 × 21600) = 0.063 N/mm² NEGLIGIBLE
IS 13920 Cl. 10.4 boundary elements NOT triggered

Sentry post — IS 1893 (Part 1):2016

Every value in Part B has been computed numerically in this project. The DXF drawings carry exactly this reinforcement — no drawing shows an arrangement that differs from these calculations.

Flexure equation used throughout — IS 456 Annex G-1.1(b):

Mu = 0.87 fy · Ast · d · [1 - (Ast · fy)/(b · d · fck)]	
xu = 0.87 fy Ast /(0.36 fck b)	IS 456 Cl. 38.1
xu,max/d = 0.46 for Fe500	IS 456 Cl. 38.1(f)
Mu,lim = 0.133 fck b d²	IS 456 Annex G-1.1(c)

B.1 Perimeter walls W1-W4 — 600 thk

INPUTS	t 600, clear height 3200, cover 75 earth / 40 internal, T16 d = 600 - 75 - 8 = 517 mm Blast 383 kPa (Ka 1.0); static earth+water 83.2 kPa at base → BLAST GOVERNS 4.6 : 1
FORMULA	Mp = w·Ln²/16 (fixed-fixed plastic mechanism)
SUBSTITUTE	Mp = 383 × 3.200²/16
RESULT	Mp = 245.1 kNm/m
CHECK	Mu,lim = 0.133 × 43.75 × 1000 × 517² = 1556 kNm/m ✓ singly reinforced Ast,req (fy,dyn 625, fck,dyn 43.75) = 894 mm²/m IS 456 Cl. 32.5(a) min vertical 0.0012 × 600 = 720 mm²/m IS 456 Cl. 32.5(b) min horizontal 0.0020 × 600 = 1200 mm²/m IS 3370 Pt 2 surface zone 0.35 % × 250 each face = 875 mm²/m/face T16 @ 150 c/c EACH FACE EACH WAY = 1340 mm²/m Mu = 362.8 kNm/m → UTILISATION 68 % xu = 46.3 → x/d = 0.089 < 0.46
ADOPTED	
SHEAR	V at d = 383 (1.600 - 0.517) = 414.8 kN/m tv = 414.8e3/(1000 × 517) Cl. 40.1 = 0.802 N/mm² pt = 0.259 % → tc (Table 19, M35) = 0.375 N/mm² tc,max (Table 20, M35) 3.70 > 0.802 ✓ Cl. 40.2.3 Vus = (0.802 - 0.375) × 517 = 220.8 kN/m Asv/sv = 220800/(0.87 × 500 × 517) Cl. 40.4(a) = 0.982 mm²/mm → T12 2-leg at 230; Cl. 26.5.1.5 limit min(0.75d 388, 300) = 300
ADOPTED	T12 CLOSED LINKS @ 200 c/c (Asv/sv = 1.131 ✓)
AXIAL	N = 448.15 × 2.50 + 3.2 × 0.6 × 25 = 1168 kN/m f = 1.95 N/mm² = 11 % of 0.4 fck,dyn (17.5) ✓ P-M not critical
CRACK	IS 3370 Pt 2, limit 0.2 mm; surface steel 875 < 1340 ✓

Why 600 mm — stated honestly. The wall is NOT strength-governed (68 %). 600 is set by (i) 75 mm cover, (ii) congestion — two curtains + closed links + waterstops + cast-in frames, (iii) IS 3370 crack control under sustained hydrostatic load, (iv) the EMP double curtain at 150, (v) the 1168 kN/m axial path.

B.2 Walls W6 / W7 — 400 thk (MODIFICATION M1)

FINDING F1	The stair shaft is open to atmosphere through the 2800 × 3160 roof void, the headhouse and the entry stairwell. Fill time V/(A·c) = 105/(3.3 × 340) = 0.094 s vs td 0.13–1.33 s → THE SHAFT EQUALISES TO FULL p_so. W6/W7 sit in the SAME PLANE as the 7-bar blast doors, separating a pressurised shaft from Bay 6 / Bay 8 at ambient.
WHY 200 IS IMPOSSIBLE (not a detailing problem)	d = 200 - 40 - 6 = 154 mm Mu,lim = 0.133 × 43.75 × 1000 × 154² = 138.0 kNm/m Demand Mp = 383 × 3.200²/16 = 245.1 kNm/m 138 < 245 → NO STEEL RATIO MAKES IT WORK Two-way action checked before rejection: panel 3200 × 3800, fixed 3 edges, yield line ≈ 0.7 × one-way = 172 kNm/m – still above 138.
DESIGN OF THE 400 WALL	d = 400 - 50 (shaft face) - 8 = 342 mm Mp = 383 × 3.200²/16 = 245.1 kNm/m Mu,lim = 0.133 × 43.75 × 1000 × 342² = 680.6 kNm/m ✓ Ast,req = 1400 mm²/m
ADOPTED	T20 @ 150 EF EW = 2094 mm²/m → Mu = 355.3, UTIL 69 %, x/d = 0.211
SHEAR	V at d = 383 (1.600 - 0.342) = 481.8 kN/m tv 1.409 tc 0.540 (pt 0.612 %) tc,max 3.70 ✓ Vus 297.2 kN/m; Asv/sv 1.998 → T12 4-leg at 226 Cl. 26.5.1.5 limit min(0.75d 257, 300) = 257
ADOPTED	T12 4-LEGGED LINKS @ 200 c/c
BLAST DOOR OPENING 1200 × 2100	Interrupted steel 2094 × 1.2 = 2513 mm²/face → 1256 each jamb
ADOPTED	4 No. T20 EACH JAMB EACH FACE (1257 mm²), anchored Ld 800 beyond
HEADER	400 × 1100 over 1200 clear; w = 383 × 2.1/2 = 402 kN/m M = wL²/12 = 48.2 kNm ; V = 241 kN ; τ = 0.578 N/mm²
ADOPTED	4-T20 top + 4-T20 bottom, T12 4-leg links @ 150
FRAME	Cast-in steel frame anchored into and WELDED to the cage (EMP). Door: proprietary, ≥ 7 bar, rebound-rated, gas-tight [V]

B.3 Mat foundation — 600 thk

BEARING Service 58.4 kPa (1.8 % of 3240) ; Blast 404.9 kPa (12.5 %) ✓
Bearing governs NOTHING.

CASE 1 – NET UPLIFT (COMB 102)
 $u = 4.700 \times 9.81 = 46.11$ kPa ; self $0.600 \times 25 = 15.0$
 net 31.1 kPa UP over 5000 clear
 $M_p = 31.1 \times 5.0^2/16 = 48.6$ kNm/m NOMINAL

CASE 2 – SOFT / RED-BOLE ZONE *** GOVERNS ***
 A 3.0 m band of red-bole or vesicular material removed, worst position
 $q = 404.9$ kPa (blast bearing) over a 3.0 m unsupported span
 $M = q \cdot L^2/12 = 404.9 \times 9/12 = 303.7$ kNm/m
 $d = 600 - 75 - 8 = 517$ mm
 $A_{st,req} = 1115$ mm²/m

ADOPTED T16 @ 150 EF EW = 1340 mm²/m → $M_u = 362.8$, UTILISATION 84 %

THICKNESS STUDY 500 → $A_{st,req}$ 1407 = 105 % of provided FAIL
 600 → 1115 = 84 % ADOPTED
 700 → 926 = 70 % | 800 (as drawn) ~ 57 % = 33 % unused reserve

ONE-WAY SHEAR – NOT CHECKED IN THE REPORT (CONFLICT C6)
 V at d from the soft-band edge = 404.9 (1.500 – 0.517) = 398.0 kN/m
 τ_v 0.770 | τ_c 0.375 (pt 0.259 %) | $\tau_{c,max}$ 3.70 ✓
 (no dynamic increase – IS 4991 Cl. 10.3.1.1)
 $V_{us} = 204.2$ kN/m ; $A_{sv}/sv = 0.908$ mm²/mm

ADOPTED T12 CLOSED LINKS ON A 250 × 250 GRID THROUGHOUT
 supplied $A_{sv}/sv = (4 \times 113.1)/250 = 1.810 = 2.0 \times$ required ✓
 Doubles as the spacer system between the two curtains.

PUNCHING IS 456 Cl. 31.6 – no columns/pedestals bear on the mat; headhouse walls bear on the ROOF. NOT APPLICABLE – declared, not ignored.

SLIDING Box socketed ~4.8 m into basalt, 300 annulus backfilled M15 lean.
 Not a credible mechanism – reasoning stated, no spurious factor computed.

OVERTURNING Buried box 22.0 × 6.2 × 4.7, symmetrical lateral load. NOT APPLICABLE.

SETTLEMENT Negligible on sound basalt (IS 12070 Cl. 6). Hazard = the flow contacts.

FLOTATION – THE CONSTRUCTION STAGE GOVERNS

Stage	Weight	Uplift	FoS @ (-)2.000	FoS flooded to GL
1 mat cast only	2 009 kN	6 175 kN	0.33 FAIL	0.23 FAIL
2 mat + walls, no roof	4 802 kN	6 175 kN	0.78 FAIL	0.55 FAIL
3 box complete, no backfill	7 528 kN	6 175 kN	1.22 MARGINAL	0.86 FLOATS
4 backfilled + cover	12 453 kN	6 175 kN	2.02 OK	1.41 OK

Require $FoS \geq 1.2$. ULS COMB 102 = $0.9 \times 12453/(1.5 \times 6175) = 1.21$
 Uplift acts on the REAL 133.92 m² underside, not the plate mid-surface area.
 At the M1 geometry (22000) both uplift and resistance scale with length,
 so EVERY FoS IS UNCHANGED.

MANDATORY MITIGATION – a design output, not a contractor's problem:

- Continuous dewatering from start of excavation until backfill and cover complete
- Temporary pressure-relief valves / knock-out plugs in the mat (6 No. on S-02), grouted up ONLY after backfill
- Programme the sub-structure to complete before the monsoon, or bund and positively drain with standby pumping and generator back-up
- Backfill SYMMETRICALLY, in layers

B.4 Pressure (roof) slab — 900 thk — THE GOVERNING ELEMENT

ONE-WAY ARGUMENT	Aspect $20.8/5.0 = 4.2$. IS 456 Table 26 is tabulated only to $l_y/l_x = 2.0$, so two-way action CANNOT be claimed. → $M_p \propto L_n^2$ of the 5 m width ONLY. → LENGTHENING THE BOX IS STRUCTURALLY FREE (basis of M1). Widening 5.0 → 6.0 m would increase the roof moment 44 %.		
NATURAL PERIOD	$m = 0.900 \times 2500 + 0.25 \times 2.0 \times 2000$ $0.5 E I_g = 0.5 \times 2.958e10 \times 0.900^3/12$ $f_1 = (22.373/2\pi) \cdot \sqrt{(EI/(m \cdot L_n^4))}$ T td/T = 10 to 100 → QUASI-STATIC	$= 3250 \text{ kg/m}^2$ $= 8.985e8 \text{ N} \cdot \text{m}^2/\text{m}$ $= 74.9 \text{ Hz}$ $= 13.4 \text{ ms}$	
FLEXURE	$d = 900 - 75 - 12.5$ $M_p = w \cdot L_n^2/16 = 448.15 \times 25/16$ $M_{u,lim} = 0.133 \times 43.75 \times 1000 \times 812.5^2$ Ast,req (pt 0.20 %) Cl. 26.5.2.1 min 0.12 % × 900 Cl. 26.5.2.2 max bar dia D/8 = 112	$= 812.5 \text{ mm}$ $= 700.2 \text{ kNm/m}$ $= 3841 \text{ kNm/m} \checkmark$ $= 1632 \text{ mm}^2/\text{m}$ $= 1080 \text{ mm}^2/\text{m}$ $\checkmark (T25)$	
ADOPTED	T25 @ 150 EF EW = 3272 mm ² /m Mu = 1362.4 kNm/m → UTILISATION 51 % xu = 113.0 → x/d = 0.139 < 0.46 *** x/d = 0.139 IS THE PROOF THAT $\mu = 5$ IS DEFENSIBLE ***		
SHEAR	V at support face = 448.15×2.50 V at d from face = $448.15 \times (2.50 - 0.8125)$ tv 0.931 τ_c 0.450 (pt 0.403 %) τ_c ,max 3.70 $\checkmark$ Vus 390.8 kN/m ; Asv/sv 1.106 → T12 4-leg at 409 Cl. 26.5.1.5 limit 300 GOVERNS	$= 1120 \text{ kN/m}$ $= 756.3 \text{ kN/m}$ $\checkmark$	
ADOPTED	T12 4-LEG @ 250 IN THE END 1500 EACH SIDE; T12 2-LEG @ 300 ELSEWHERE		
DIRECT SHEAR	Vd,max = 0.16 fck,dyn b d vs 1120 applied → 19.7 %	$= 5688 \text{ kN/m}$ [UFC 3-340-02 §4-30, NOT IS 456]	
THICKNESS STUDY	t d Mp Ast,req tv verdict 600 512.5 689 2673 1.71 links at 164, congested 800 712.5 696 1868 1.12 structural optimum 900 812.5 700 1633 0.93 ADOPTED 1000 912.5 704 1453 0.78 11 % over 900 adopted for (a) reserve at the 1400 collars, (b) headroom against a raised DBT while the yield is unconfirmed, (c) support rotation inside 2° without UFC lacing.		

B.4.1 Roof openings

(a) Two 1400 dia circular escape-shaft openings

Interrupted steel = 3272×1.400 = 4581 mm²/face/direction
 Trimmers each side = 4581/2 = 2291 mm²
 ADOPTED 5 No. T25 EACH SIDE, EACH FACE, EACH DIRECTION (2454 mm²)
 anchored Ld = 1000 beyond the opening
 Collar 250 RC; T16 @ 150 hoops (3 layers) + T16 @ 150 radials
 Slab thickened 900 → 1200 over a 600 mm annulus
 Circular is the right shape: hoop action, NO re-entrant stress concentration.
 This detail SETS the 750 mm clearance rule.

(b) The 2800 × 3160 rectangular stair void — FINDING F2, not designed in the report

The void leaves an 1840 mm CANTILEVER of 900 slab, 2800 wide, at 448.15 kPa.
 $M(\text{root}) = 448.15 \times 1.840^2/2$ = 758.6 kNm/m
 *** THIS EXCEEDS THE 700.2 kNm/m MIDSPAN M_p ***
 $A_{st,req} \text{ TOP} = 1772 \text{ mm}^2/\text{m} < 3272 \text{ provided}$ → UTILISATION 56 % $\checkmark$
 $V(\text{root}) = 448.15 \times 1.840$ = 824.6 kN/m
 $\tau_v 1.015 > \tau_c 0.450$; Vus 459.4 kN/m ; Asv/sv 1.300 → T12 4-leg at 348
 Cl. 26.5.1.5 limit 300
 ADOPTED T12 4-LEG LINKS @ 250 THROUGHOUT THE PAD (NEW requirement)
 plus Free edge thickened 900 → 1200 over 600, with 6-T25 top + 6-T25 bottom
 and T12 closed links @ 150
 plus 6 No. T25 each face, top and bottom, in a 900 band over W6 and W7
 plus Diagonal trimmers 4 No. T25 each face at 45° at BOTH re-entrant corners,
 2000 long each way, anchored Ld beyond
 plus 1100 mm guarding to the free edge (NBC 2016 Pt 4)

B.5 Main staircase — Bay 7

STATUS: INSIDE the protective envelope. NOT a blast element. The shaft pressurises, but that pressure acts on the shaft WALLS and the blast DOORS — not as a net load on a slab open on both faces. **Designed to IS 456 with normal partial factors, NOT with IS 4991 dynamic strengths.** Stated explicitly on drawing S-04.

Geometry check — NBC 2016 Part 4

Item	Provided	Limit	Verdict
Riser	170.833 mm	≤ 190	✓
Tread / going	280 mm	≥ 250	✓
Flight width	1200 mm	≥ 1000	✓ stretcher-capable
Risers per flight	8	≤ 12 preferred	✓
Headroom	2533 mm	≥ 2200	✓
Total rise	4100 = 24 × 170.833	(-)6.100 → (-)2.000	closes exactly

```
FLIGHT – waist 200
  θ = arctan(170.833/280) = 31.4°, cos θ = 0.8535
  waist self 0.200 × 25/0.8535 = 5.858 kPa
  steps 0.5 × 0.170833 × 25 IS 456 Cl. 33.2 = 2.135 kPa
  finishes = 1.000 kPa
  live IS 875 (Pt 2), plant + escape route = 5.000 kPa
  total 13.99 → wu = 1.5 × 13.99 = 21.0 kPa
  Leff = going + min(landing/2, 1000) each end IS 456 Cl. 33.1(b)
    = 1960 + 600 + 600 = 3160 mm
  d = 200 – 30 – 6 = 164 mm
  M = wu·Leff²/8 = 26.2 kNm/m
  Ast,req 380 ; Cl. 26.5.2.1 min 0.12 % × 200 = 240
  ADOPTED T12 @ 150 MAIN (754) → Mu 50.3, UTIL 52 % ; T10 @ 200 DISTRIBUTION (393)
  DEFLECTION Leff/d = 19.3 ; basic 20 ; fs = 0.58 × 500 × 380/754 = 146 ; pt 0.46 %
    MF ≈ 1.9 → permissible 38 > 19.3 ✓
  SHEAR V = 33.2 kN/m ; tv 0.202 ; tc 0.479 × k 1.20 (Cl. 40.2.1.1) = 0.575 ✓ no links
  TOP STEEL T12 @ 150 for 0.25 Leff = 800 into the flight, anchored Ld 480

LANDINGS L1 / L2 / ARRIVAL – 200 thk, spanning ACROSS the shaft
  self 7.50 + finishes 1.50 + live 7.50 + flight reaction 27.70 = 44.20 kPa
  Leff = 2800 clear + 200 bearing Cl. 22.2(a) = 3000 mm
  M = 44.2 × 3.000²/8 = 49.7 kNm/m
  Ast,req 745 ; T12 @ 150 gives 754 = 100 % T00 TIGHT
  ADOPTED T12 @ 125 BOTTOM (905) → Mu 59.5, UTIL 84 %
    T12 @ 125 TOP for 900 from each support ; T10 @ 200 distribution
  DEFLECTION 3000/164 = 18.3 ; fs 239 ; pt 0.552 % ; MF ≈ 1.35 → 27 > 18.3 ✓
  SHEAR V 66.3 kN/m ; tv 0.404 < tc 0.519 × 1.20 = 0.622 ✓
```

Compatibility notes: the arrival landing at (-)6.100 IS the mat surface, no separate slab. Flight 3 lands directly on the 900 pressure-slab pad at (-)2.000. The landings prop W6/W7 at (-)4.733 and (-)3.367 **but only over their 1200 depth — the 400 wall design in B.2 does NOT rely on that prop.**

B.6 Entry (approach) stairwell

STATUS: OUTSIDE the protective boundary, NOT blast rated (Rev F drawing note 8). Expected to be LOST in the design event.
Designed to IS 456 with normal partial factors.

FLIGHT – 12R @ 166.6667 / 300, 1500 wide, waist 250
 $\theta = 29.05^\circ$, $\cos \theta = 0.8742$
waist 7.150 + steps 2.083 + finishes 1.000 + live 5.000 = 15.233 → wu = 22.85 kPa
Leff = 3300 + 750 + 750 IS 456 Cl. 33.1(b) = 4800 mm
d = 250 – 30 – 8 = 214 mm
M = 22.85 × 4.800²/8 = 65.8 kNm/m
Ast,req 744 ; min 0.12 % × 250 = 300
ADOPTED T16 @ 200 MAIN (1005) → Mu 87.3, UTIL 75 %, x/d 0.162
T10 @ 200 DISTRIBUTION (393)
T16 @ 200 TOP for 1200 into the flight, anchored Ld 640
DEFLECTION 4800/214 = 22.4 ; basic 20 ; fs 215 ; pt 0.470 % ; MF ≈ 1.5 → 30 ✓
SHEAR V 54.8 kN/m ; tv 0.256 ; tc 0.484 × k 1.10 = 0.532 ✓ no links

SIDE WALLS 250 – retained 2.9 m, propped by roof and raft
oh at base = 0.5 × 20 × 2.900 + 0.5 × 10 = 34 kPa
M ≈ w·L²/12 on an equivalent 20 kPa UDL = 14.0 kNm/m
d = 250 – 50 – 6 = 194 ; Ast,req 168
IS 456 Cl. 32.5(a) min 0.0012 × 250 = 300 ; Cl. 32.5(b) 0.0020 × 250 = 500
Cl. 32.5(c) two curtains required (t > 200) ✓
ADOPTED T12 @ 200 EF EW (565/face) – MINIMUM STEEL GOVERNS, 31 % utilised
SHEAR V ≈ 49 kN/m ; tv 0.253 < tc 0.484 ✓

RAKING ROOF 250 – 1500 clear
self 6.25 + wp/screed 2.0 + earth lap 5.4 + IMPOSED 20 = 33.65 → wu 50.5 kPa
(20 kPa imposed deliberately, for a stray vehicle on the berm)
M = w·L²/12 = 9.5 kNm/m
d = 250 – 30 – 6 = 204 ; Ast,req 108 ; min 300 GOVERNS
ADOPTED T12 @ 200 EF EW → Mu 48.2, 20 % utilised
SHEAR V 38 kN/m ; tv 0.186 ✓

TOP LANDING 250 (0.000) wu 54.9 kPa ; Leff 1750 ; M 21.0 kNm/m ; Ast,req 229
ADOPTED T12 @ 200 EF EW (565) → Mu 50.6, 41 % utilised
PLATFORM 250 ((-)2.000, on fill) ADOPTED T12 @ 200 EF EW, nominal
HEADWALL 250 ADOPTED T12 @ 200 EF EW
Door lintel over 1000 clear: 250 × 350, 3-T12 top + 3-T12 bottom, T8 @ 150
STEPPED RAFT 300 on compacted fill ADOPTED T12 @ 200 EF EW

DETAILING – THE OPENING CORNER (the detail most often got wrong)
At the TOP of the flight the tension face turns through 209°. A bar bent round that corner has its bend resultant directed OUT of the concrete: it spalls the cover and the bar loses anchorage.
MAIN BARS SHALL NOT BE BENT ROUND IT. Each layer continued straight, CROSSED, anchored Ld = 640 into the OPPOSITE face, plus a U-bar T16 @ 200 across the corner.
[SP 34:1987 Cl. 5.5 and standard detailing practice]
At the FOOT of the flight the corner CLOSES (151°) – bars may turn normally.

B.7 Entry headhouse — walls and roof

B.7.1 Roof, 500 thk — three analyses, most conservative adopted

w = 383 (blast) + 12.5 (self) + 1.0 (SIDL) = 396.5 kPa
Panel 4000 × 5000 clear, monolithic with 400 walls on all four sides, ly/lx = 1.25
d = 500 – 75 – 10 = 415 mm
Mu,lim = 0.133 × 43.75 × 1000 × 415² = 1002 kNm/m

#	Method	M
1	One-way fixed-fixed strip, Mp = w·Ln ² /16	396.5 ← ADOPTED
2	IS 456 Table 26 Case 1, α _x = 0.045 at 1.25	285.5
3	Yield line, FIXED 4 edges, isotropic	162.2

Method 3 formula: wu = 48m / [a²(√(3 + (a/b)²) – a/b)²]
VALIDATION: as b → ∞, 48/3 = 16 = the fixed-fixed strip w·L²/16 ✓
(the 24-coefficient version gives 8 = SIMPLY SUPPORTED w·L²/8)

Ast,req (one-way basis) = 1879 mm²/m
Cl. 26.5.2.1 minimum 0.12 % × 500 = 600 mm²/m
ADOPTED T20 @ 150 EF EW = 2094 mm²/m → Mu = 438.5 kNm/m
UTILISATION 90 % one-way | 65 % Table 26 | 37 % yield line
xu 72.3 → x/d = 0.174 << 0.46

SHEAR *** NEW AND MANDATORY – NOT IN THE PHASE 1 REPORT ***
V at d = 396.5 (2.000 – 0.415) = 628.4 kN/m
tv 1.514 | tc 0.502 (pt 0.505 %) | tc,max 3.70 ✓
(tc read at the STATIC M35 value – IS 4991 Cl. 10.3.1.1)
Vus 420.0 kN/m ; Asv/sv 2.327 → T12 4-leg at 194
Cl. 26.5.1.5 limit min(0.75d 311, 300) = 300
ADOPTED T12 4-LEG @ 175 IN THE END 1200 EACH SIDE; @ 250 ELSEWHERE

DETAILING No opening in the headhouse roof (the stair void is in the FLOOR).
Corner torsion steel (Cl. D-1.8) NOT required – all four edges carry full T20 @ 150 hogging steel continuous into the walls.
Haunch 300 × 300 at all four wall-roof junctions, diagonal T16 @ 150.
Top steel continuous over every support, anchored Ld = 800 into the wall.

B.7.2 Walls HW1-HW4, 400 thk — CONFLICT C10

	CASE A (report)	CASE B (ADOPTED)
Lateral pressure	113 kPa drag	383 kPa full envelope
$M_p = w \times 2.400^2/16$	40.7 kNm/m	137.9 kNm/m
Ast,req	221 mm ² /m	766 mm ² /m
Cl. 32.5(a) min vertical	480 mm ² /m – governs	480
Cl. 32.5(b) min horizontal	800 mm ² /m – governs	800
Utilisation on T16@150 EF EW	17 %	59 %
Shear V at d	97.0 kN/m	328.6 kN/m
τ_v	0.283	0.961
τ_c (pt 0.392 %)	0.444	0.444
Verdict	NO LINKS	LINKS REQUIRED

d = 400 – 50 – 8 = 342 ; $M_{u,lim}$ = 681 kNm/m
ADOPTED T16 @ 150 EF EW = 1340 mm²/m → M_u 235.2, UTIL 59 %, x/d 0.135
Vus 176.9 kN/m ; Asv/sv 1.189 → T12 4-leg at 381
Cl. 26.5.1.5 limit min(0.75d 256, 300) = 256 GOVERNS
ADOPTED T12 4-LEGGED LINKS @ 250 c/c THROUGHOUT ALL FOUR WALLS

Two-way (information only, fixed 4 edges):
HW1/HW2 4000 × 2400 → m = 69.9 kNm/m (one-way strip 137.9)
HW3/HW4 5000 × 2400 → m = 79.8 kNm/m (one-way strip 137.9)

AXIAL Roof reaction, two-way (Cl. 24.5): HW3/HW4 trapezoid 2379 kN / 5.0 m = 476 kN/m
HW1/HW2 triangle 1586 kN / 4.0 m = 396 kN/m
One-way bound: 396.5 × 2.0 = 793 kN/m ; self 29 kN/m
Worst N = 822 kN/m → f = 2.06 N/mm² = 12 % of 0.4 f_{ck,dyn} ✓ not critical

PRESSURE ACTS FROM INSIDE T00. The inner security door is NOT blast rated and the entry stairwell is expected to be lost, so the headhouse fills and the walls are pushed OUTWARDS. *** THE WALLS ARE REINFORCED SYMMETRICALLY FOR 383 kPa EITHER WAY – a stated requirement, not an accident of detailing. ***

DOOR OPENING 900 × 2100 IN HW2 – and a geometric trap
Interrupted steel 1340 × 0.900 = 1206 mm²/face → 603 each jamb
ADOPTED 2 No. T20 EACH JAMB EACH FACE (628), anchored Ld 800 beyond
Door head at (-)2.000 + 2.100 = +0.100 ; roof soffit +0.400
→ ONLY 300 mm OF WALL ABOVE THE DOOR – too shallow for a lintel.
The 300 wall and the 500 roof act TOGETHER as an 800 mm deep edge band:
w = 383 × 2.100/2 = 402 kN/m ; M = wL²/12 = 27.1 kNm ; V = 181 kN
d = 800 – 40 – 8 – 10 = 742 ; τ_v = 0.610 N/mm²
ADOPTED 400 × 800 EDGE BAND: 4-T20 TOP + 4-T20 BOTTOM, T12 4-LEG @ 150,
over the opening and 600 each side
Door frame cast in and WELDED to the cage (EMP), even though the leaf is a security door and not blast rated.

B.7.3 HW3 – the wall that has nothing under it

The pressure slab spans one-way in Y over 5000 clear. HW3 runs IN the Y direction, so its load is carried by a slab strip acting as a beam.

$$b_{eff} = 0.400 + 2 \times 1.05 \text{ (45° spread through the 900 slab)} = 2.5 \text{ m [R]}$$

CASE 1 – two-way roof distribution (IS 456 Cl. 24.5, correct)
Line load 476 (trapezoid) + 29 (self) = 505 kN/m
Equivalent UDL = 505/2.5 + 22.5 + 2.0 = 226 kPa
M = w·Ln²/16 = 226 × 25/16 = 354 kNm/m
Against the slab's 1362 kNm/m capacity = 26 % ✓

CASE 2 – one-way roof bound
Line load 793 + 29 = 822 kN/m → 353 kPa → M = 552 kNm/m = 41 % ✓

ADOPTED No slab change needed. Detailed as a LINE LOAD, not as a support.
Robustness: 4 No. T25 ADDITIONAL TOP AND BOTTOM in a 1200 mm band beneath HW3, lapped 1250 (50φ) into the main mesh.

B.8 Sentry post

NOT designed for the blast, and that is a RECORDED decision. Reflected force on the 4.0 × 6.7 m face = 1366 kPa × 26.8 m² = **36 600 kN (≈ 3730 t)**. Hardening would need ~700 mm of RC on all four above-ground faces. Not justified for a peacetime observation post. **All member design uses the STAAD base shear V_b = 73.18 kN, not the hand-calculated 59.3 kN.**

B.8.1 Seismic member forces – portal method at V_b = 73.18 kN

Q_{roof} = 73.18 × (315.9 × 6.25²)/Σ = 59.51 kN
Q_{floor} = 73.18 × (276.8 × 3.20²)/Σ = 13.67 kN
Column shear, ground storey = 73.18/4 (= STAAD Fx 18.295) ✓ = 18.30 kN
Column shear, first storey = 59.51/4 = 14.88 kN
Column moment, ground = 18.30 × 3.200/2 = 29.27 kNm
Column moment, first = 14.88 × 3.050/2 = 22.69 kNm
Beam moment, first-floor joint (corner joint, one beam) = 29.27 + 22.69 = 51.96 kNm
Beam moment, roof joint = 22.69 kNm
Axial from overturning – READ DIRECTLY FROM THE STAAD REACTIONS:
EQ+X Fy = ± 36.127 kN EQ+Z Fy = ± 27.891 kN → 36.127 governs

B.8.2 Frame analysis assumption – stated, not hidden

Single-bay portal each direction. Beams are NOT fully fixed; the joint rotates.
 Slope-deflection with symmetric loading:
 $I_{\text{beam}} = 250 \times 450^3 / 12 = 1.8984\text{e}9 \text{ mm}^4$; $I_{\text{col}} = 350^4 / 12 = 1.2505\text{e}9 \text{ mm}^4$
 $K_c \text{ (first-floor joint)} = 4EI/3200 + 4EI/3050 = 3.203\text{e}6 \cdot E$
 $M_{\text{support}} = -FEM + (2EI_b/L_b) \cdot \theta$, $\theta = FEM / (2EI_b/L_b + K_c)$

Beam	FEM	2EI/L	Reduction	M_support	M_midspan
B1, 1.5(DL+LL)	39.49	1.040e6 E	$\times 0.755$	29.81	30.90
B1, DL only	22.53	—	$\times 0.755$	17.01	—
B2, 1.5(DL+LL)	70.60	0.8165e6 E	$\times 0.797$	56.26	49.64
B2, DL only	39.21	—	$\times 0.797$	31.25	—

B.8.3 Slab S1 — 150 thk two-way

Cover 30 (Table 16, moderate) ; T8 bars
 $dx = 150 - 30 - 4 = 116 \text{ mm}$ $dy = 150 - 30 - 8 - 4 = 108 \text{ mm}$
 Clear spans 3400 and 4400
 Effective span, IS 456 Cl. 22.2(a) – lesser of clear + d, or c/c:
 $l_x = \min(3400 + 116, 3650) = 3516$ $l_y = \min(4400 + 108, 4650) = 4508$
 $l_y/l_x = 1.282 < 2 \rightarrow \text{TWO-WAY}$ Cl. 24.4 / Annex D

MOMENT COEFFICIENTS – IS 456 Table 26 (Annex D-1.1) Case 9, four edges
 discontinuous but RESTRAINED, with corner torsion steel per Cl. D-1.8
 $\rightarrow$ Table 26, NOT Table 27.
 $\alpha_x = 0.0777$ (interpolated at 1.282) $\alpha_y = 0.056$

$w_u = 1.5(4.750 + 3.000) = 11.625 \text{ kPa}$ (the FIRST FLOOR governs; roof = 10.125)
 $M_x = 0.0777 \times 11.625 \times 3.516^2 = 11.17 \text{ kNm/m}$
 $M_y = 0.0560 \times 11.625 \times 3.516^2 = 8.05 \text{ kNm/m}$
 $A_{st, \text{req } x} = 229$; $y = 176$; Cl. 26.5.2.1 min 0.12 % $\times 150 = 180 \text{ mm}^2/\text{m}$
 Cl. 26.5.2.2 max bar dia $D/8 = 18.75 \checkmark$; Cl. 26.3.3(b) max spacing $3d$ or $300 \checkmark$
 ADOPTED T8 @ 150 c/c BOTH WAYS BOTTOM = $335 \text{ mm}^2/\text{m}$
 $\mu_u = 16.09 \text{ (x)} / 14.92 \text{ (y)} \text{ kNm/m} \rightarrow \text{UTILISATION } 69 \% / 54 \%$
 $x_u = 13.5 \rightarrow x/d = 0.116 / 0.125$

DEFLECTION $l_x/dx = 3516/116 = 30.3$; basic taken as SIMPLY SUPPORTED = 20
 $f_s = 0.58 \times 500 \times 229/335 = 198$; $p_t = 0.289 \%$; MF (Fig. 4) ≈ 1.68
 Permissible $20 \times 1.68 = 33.6 > 30.3 \checkmark$
 (as continuous, basic = 26 and the margin doubles – it passes on the conservative assumption, so that assumption is used)

SHEAR $V_u \approx w_u \cdot l_x/3 = 13.6 \text{ kN/m}$; $\tau_v 0.117$; $\tau_c 0.390 \times k 1.30 \text{ (D} \leq 150) = 0.507 \checkmark$

DETAILING – IS 456 ANNEX D
 Cl. D-1.2/D-1.3 middle strip = $\frac{3}{4}$ width, edge strips = $\frac{1}{8}$ each side;
 Table 26 moments apply to MIDDLE STRIPS ONLY
 Cl. D-1.4 50 % of midspan bottom steel extends to within 0.1 L of a discontinuous edge; remainder curtailed at 0.25 L
 Cl. D-1.6 AT EVERY DISCONTINUOUS EDGE provide top steel = 50 % of the midspan bottom steel, extending 0.1 L = 400 mm into the span
 $\rightarrow$ ADOPTED T8 @ 300 TOP, 400 wide band, all four edges
 Cl. D-1.7 edge strips: minimum reinforcement only
 Cl. D-1.8 TORSION AT CORNERS – both edges discontinuous at all four corners $\rightarrow$ steel = $\frac{3}{4}$ of the midspan area, in FOUR layers (2 top + 2 bottom, each way), over $l_x/5 = 700 \text{ mm}$ each way
 $\rightarrow$ ADOPTED T8 @ 200, 4 LAYERS, 700×700 AT EACH CORNER
 *** THE TORSION MATS ARE A HOLD POINT FOR REINFORCEMENT INSPECTION. Table 26 is valid only because they are provided. Omit them and the slab must be reassessed on Table 27. ***

B.8.4 Beam B1 — 250 × 450, span 3650

IS 13920:2016 section limits: $b \geq 200 \checkmark$ (250) | $b/D = 0.556 \geq 0.3 \checkmark$ |
 $\text{clear span}/D = 3300/450 = 7.33 \geq 4 \checkmark$
 $d = 450 - 30 - 8 - 8 = 404$; $M_u, \text{lim} = 0.133 \times 30 \times 250 \times 404^2 = 162.8 \text{ kNm}$

Combination	M_support hogging
1.5(DL + LL)	29.8
1.2(DL + LL + EL) = 1.2(19.87 + 51.96)	86.2
1.5(DL + EL) = 1.5(17.01 + 51.96)	103.5 ← GOVERNS
0.9 DL + 1.5 EL	93.2

$M_u = 103.5 < 162.8 \rightarrow$ singly reinforced ; $A_{st, \text{req}} = 661 \text{ mm}^2$
ADOPTED 4-T16 TOP AT SUPPORTS (804) $\rightarrow M_u = 122.6 \text{ kNm}$, UTIL 84 %, x/d 0.321

BOTTOM Reversal $1.5(17.01 - 51.96) = -52.4 \text{ kNm}$ sagging $\rightarrow A_{st, \text{req}} 313$
IS 13920 Cl. 6.2.3: positive steel at joint face ≥ 50 % of negative
 $= 0.5 \times 804 = 402 \text{ mm}^2 \leftarrow$ GOVERNS
Midspan sagging $30.9 \text{ kNm} \rightarrow A_{st, \text{req}} 181$
IS 13920 Cl. 6.2.1(b) $p_{\text{min}} = 0.24/f_{ck}/f_y = 0.00263 \times 250 \times 404 = 266$
IS 456 Cl. 26.5.1.1 $A_{s, \text{min}} = 0.85b_d/f_y = 172$
IS 13920 Cl. 6.2.2 $p_{\text{max}} = 0.025 \rightarrow 2525 \checkmark$
Cl. 6.2.4 top or bottom ≥ 25 % of max at joint face = 201 $\checkmark$
ADOPTED 2-T16 BOTTOM CONTINUOUS (402) $\rightarrow M_u = 66.0 \text{ kNm}$, x/d 0.160

SHEAR – capacity design, IS 13920 Cl. 6.3.3
 $V_u = 1.2(D+L) \cdot L/2 \pm 1.4(M_u^{\text{As}} + M_u^{\text{Bh}})/L$
Unfactored gravity total = 54.29 (UDL) + 25.81 (triangle) = 80.1 kN
 $1.2 \times 80.1/2 = 48.1$; $1.4 \times (122.6 + 66.0)/3.650 = 72.3$
 $V_u = 120.4 \text{ kN}$; $\tau_v = 1.192 \text{ N/mm}^2$; $\tau_{c, \text{max}}$ (Table 20, M30) $3.50 \checkmark$
*** IS 13920 Cl. 6.3.4: EQ share $72.3/120.4 = 60$ % ≥ 50 %, no axial
 $\rightarrow \tau_c$ TAKEN AS ZERO in the plastic hinge region ***
 $A_{sv}/s_v = 120.4e3/(0.87 \times 500 \times 404) = 0.685 \rightarrow$ T8 2-leg at 147
Cl. 6.3.5.1 hinge region ($2d = 810$ from each face):
 $s_v \leq \min(d/4 = 101, 8 \times 16 = 128)$, need not be $< 100 \rightarrow 100 \text{ mm}$
Cl. 6.3.5.2 elsewhere: $s_v \leq d/2 = 202$
ADOPTED T8 2-LEGGED HOOPS @ 100 c/c OVER $2d = 810$ FROM EACH FACE;
T8 @ 150 c/c ELSEWHERE. First hoop within 50 mm of the column face.
DEFLECTION $l/d = 3650/404 = 9.0 \ll 20 \checkmark$

B.8.5 Beam B2 — 250 × 450, span 4650

Combination	M_support hogging
1.5(DL + LL)	56.3
1.2(DL + LL + EL) = 1.2(37.51 + 51.96)	107.4
1.5(DL + EL) = 1.5(31.25 + 51.96)	124.8 ← GOVERNS
0.9 DL + 1.5 EL	106.1

$M_u = 124.8 < 162.8 \checkmark$; $A_{st, \text{req}} = 822 \text{ mm}^2$
ADOPTED 3-T20 TOP AT SUPPORTS (942) $\rightarrow M_u = 139.8 \text{ kNm}$, UTIL 89 %, x/d 0.376
*** 3-T20 chosen over 5-T16: five T16 need 256 mm in a 250 wide beam ***
BOTTOM Cl. 6.2.3 $\rightarrow 0.5 \times 942 = 471$; midspan $49.6 \text{ kNm} \rightarrow A_{st, \text{req}} 297$
ADOPTED 2-T20 BOTTOM CONTINUOUS (628) $\rightarrow M_u = 98.9 \text{ kNm}$, 50 % utilised, x/d 0.250

SHEAR Unfactored gravity (UDL-equivalent) = $26.115 \times 4.650 = 121.4 \text{ kN}$
 $1.2 \times 121.4/2 = 72.8$; $1.4 \times (139.8 + 98.9)/4.650 = 71.9$
 $V_u = 144.7 \text{ kN}$; $\tau_v = 1.433$; $\tau_{c, \text{max}} 3.50 \checkmark$
EQ share $71.9/144.7 = 49.7$ % – a hair under the Cl. 6.3.4 50 % trigger.
*** TAKEN AS TRIGGERED ($\tau_c = 0$): a design must not turn on the third decimal place of a ratio. ***
 $A_{sv}/s_v = 0.823 \rightarrow$ T8 2-leg at 122 ; hinge spacing 100 governs
Outside the hinge, V falls to $\sim 124 \text{ kN}$ and $\tau_c = 0.641$ (pt 0.933 %) may be used
 $\rightarrow A_{sv}/s_v 0.337 \rightarrow$ T8 at 298
ADOPTED T8 @ 100 OVER 810 FROM EACH FACE ; T8 @ 150 ELSEWHERE
DEFLECTION $l/d = 4650/404 = 11.5 \ll 20 \checkmark$

B.8.6 Column C1 — 350 × 350

SLENDERNESS clear height ground = 3650 - 450 - 450 (beam depth) = 2750 mm
Effective length factor 1.2 (Table 28, unbraced, both ends restrained)
 $l_{ex} = 3300$; $l_{ex}/D = 9.43 < 12 \rightarrow$ SHORT COLUMN IS 456 Cl. 25.1.2 ✓

MIN ECCENTRICITY $e_{min} = l/500 + D/30 = 5.5 + 11.67 = 17.2$, but $\geq 20 \rightarrow 20$ mm
 $M_{min} = 276.8 \times 0.020 = 5.5$ kNm $\ll$ applied ✓ IS 456 Cl. 25.4

AXIAL Per column unfactored: DL 148.4 kN, LL 19.1 kN ; EQ axial ± 36.127 kN
 $1.5(DL + LL) = 251.3$; $1.5(DL + EL) = 276.8 \leftarrow \text{max}$; $1.5(DL - EL) = 168.4 \leftarrow \text{min}$

LONGITUDINAL STEEL IS 456 Cl. 26.5.3.1: min 0.8 % = 980 ; max 6 % = 7350
ADOPTED 8-T16 = 1608 mm² $\rightarrow p = 1.31$ % ; bars on all four faces
Cover 40 (Cl. 26.4.2.2) ; $d' = 40 + 8 + 8 = 56$; $d'/D = 0.16$

UNIAXIAL CAPACITY – computed from the IS 456 Cl. 38.1 stress block and the Fe500 design curve (Fig. 23B), NOT read off a chart:

Pu (kN) Mu1 (kNm)
168.4 105.8
251.3 111.7
276.8 112.9

BIAXIAL – IS 456 Cl. 39.6
 $P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc} = 0.45 \times 30 \times 120892 + 0.75 \times 500 \times 1608 = 2235$ kN
 $P_u/P_{uz} = 276.8/2235 = 0.124 < 0.2 \rightarrow \alpha_n = 1.0$
Worst case, EQ in Z with 30 % in X (IS 1893 Cl. 6.3.2.2):
 $M_{ux} = 25.6$ kNm ; $M_{uy} = 66.8$ kNm
 $25.6/112.9 + 66.8/112.9 = 0.227 + 0.592 = 0.819 \leq 1.0$ ✓

STRONG COLUMN – WEAK BEAM – IS 13920 Cl. 7.2.1, $\Sigma M_c \geq 1.4 \Sigma M_b$

Joint	ΣM_b	1.4 ΣM_b	ΣM_c	Verdict
First floor, X (B1)	122.6	171.6	112.9+101=213.9	✓
First floor, Z (B2)	139.8	195.7	213.9	✓
Roof, Z (B2 roof, 2-T20)	98.9	138.5	101	✓ marginal

TRANSVERSE REINFORCEMENT
General, IS 456 Cl. 26.5.3.2: tie dia $\geq \max(6, \phi/4) = 8$; spacing $\leq \min(350, 16\phi = 256, 300) = 256$
IS 13920 Cl. 7.4: hoops over the full length at $\leq$ half the least dimension = 175
ADOPTED T8 HOOPS @ 150 c/c outside the confining zone

SPECIAL CONFINING REINFORCEMENT – IS 13920 Cl. 8.1
 $l_o \geq \max(350, \text{clear height}/6 = 458, 450) = 500$ mm
spacing $\leq \frac{1}{4} \times 350 = 87.5$, but $75 \leq s \leq 100 = 85$ mm
 $A_{sh} = 0.18 s \cdot h \cdot (f_{ck}/f_y) \cdot [(A_g/A_k) - 1]$
 $h = 350 - 2 \times 40 = 270$; $A_g = 122500$; $A_k = 270^2 = 72900$
 $= 0.18 \times 85 \times 270 \times (30/500) \times (1.680 - 1) = 168.5$ mm²
minimum = $0.05 \times 85 \times 270 \times 30/500 = 68.9$ mm²
ADOPTED T10 HOOPS + ONE CROSS-TIE EACH WAY @ 85 c/c (3 legs $\times 78.5 = 235.5$ ✓)
over 500 mm from every joint face, top and bottom of every column,
AND THROUGH THE JOINT (Cl. 8.2)

DRIFT – IS 1893 Cl. 7.11.1: limit $0.004 \times \text{storey height} = 12.8$ mm (ground),
12.2 mm (first). Calculated drift under 3 mm. ✓ Not critical, but checked.

B.8.7 Isolated footing F1 – 1500 × 1500 × 600 on in-situ basalt at (–)2.000

> Founded on IN-SITU ROCK, never on backfill. This is the whole reason the sentry
> post was relocated ≥ 10 m clear of the shelter excavation.

SERVICE $P = 167.5$ (col) + 36.1 (EQ axial) + 33.75 (footing self) = 237.4 kN
M (base, EQ) = 29.3 kNm
 $e = 29.3/237.4 = 0.123$ m $< L/6 = 0.250 \rightarrow$ NO TENSION ✓
 $q_{max} = P/A(1 + 6e/L) = 105.5 \times 1.494 = 157.6$ kPa
vs presumptive SBC 3240 kPa (IS 1904 Table 1) [A] = 4.9 % ✓

ULS Governing $1.5(DL + EL)$: $P_u = 1.5(148.4 + 36.127) = 276.8$ kN
 $M_u = 1.5 \times 29.27 = 43.9$ kNm
 $e_u = 0.128$; $q_{u,max} = 276.8/2.25 \times (1 + 6 \times 0.128/1.5) = 190.0$ kPa

FLEXURE Cantilever from the column face = $(1500 - 350)/2 = 575$ mm
 $M_u = 190.0 \times 0.575^2/2 = 31.4$ kNm/m
 $d = 600 - 50$ (Cl. 26.4.2.1) – 8 = 542 mm
 $A_{st,req} 136$; Cl. 26.5.2.1 min 0.12 % $\times 600 = 720 \leftarrow$ GOVERNS

ADOPTED T12 @ 150 c/c BOTH WAYS BOTTOM (754 mm²/m), $x/d 0.056$

ONE-WAY SHEAR IS 456 Cl. 34.2.4.1, critical at d from the column face:
 $575 - 542 = 33$ mm from the edge ; $V_u = 6.3$ kN/m ; $\tau_v = 0.012$ ✓

PUNCHING IS 456 Cl. 31.6.1, critical perimeter at $d/2$:
 $b_0 = 4 \times (350 + 542) = 3568$; $V_u = 276.8 - 190.0 \times 0.892^2 = 125.6$ kN
 $\tau_v = 0.065$; $k_s = (0.5 + \beta_c) \leq 1.0 = 1.0$;
 $\tau_c = 0.25\sqrt{30}$ (Cl. 31.6.3.1) = 1.369 $\gg 0.065$ ✓

ANCHORAGE Column starters T16: $L_d, \text{compression} = 37 \times 16 = 592$ mm
Available $600 - 50 - 24 = 526$ straight + 8 ϕ bend 128 = 654 ✓

*** THE 600 mm DEPTH IS GOVERNED BY THE ANCHORAGE OF THE COLUMN STARTERS, NOT BY BENDING OR SHEAR – stated so that nobody "optimises" it to 350. ***

PART C — WHAT IS NOT DEMONSTRATED

This section must be reproduced in every future revision. It is the boundary of the design's validity.

Demonstrated to IS 456: ultimate flexural capacity, diagonal-tension shear, minimum and maximum steel, bar spacing, cover, development and anchorage, deflection, crack-width provisions, biaxial column interaction, punching and one-way shear in the footing, and ductile detailing to IS 13920 for the sentry post frame.

NOT demonstrated, and NOT claimed:

1. **Support rotation / ductility ratio for the blast case.** $\theta \leq 2^\circ$ corresponds to $\delta \leq 87 \text{ mm} = \text{span}/57$ and requires a **non-linear SDOF check** (IS 4991 Fig. 6 / Biggs) or explicit non-linear FE with strain-rate-dependent materials. **Phase 3.**
2. **Shock-wave propagation down the entry shaft** and the resulting door loading — CFD or shock tube [V].
3. **Transient soil-structure interaction** — participating mass, arching, interface pressure ratios.
4. **Blast door, valve and hatch performance** — vendor-tested [V], not a civil deliverable.
5. **Global flotation cannot be shown by an elastic-mat model** — the springs take tension, so in the model the mat never lifts. It is a hand check on real dimensions (B.3). *Anyone who says “the analysis shows no uplift” has misunderstood their own model.*
6. **The berm's actual pressure-transmission factor K_a** — removed from the critical path by designing the headhouse walls to the upper bound (C10).

A conventional static STAAD.Pro analysis gives the correct DEMAND. It is not, and must never be presented as, proof of blast resistance. IS 4991:1968 Cl. 1.1 expressly EXCLUDES nuclear explosions from its scope. It is used here for its loading rules and dynamic material strengths only, as a documented conservative extrapolation. Every structural capacity is computed to IS 456.

PART D — STAAD.PRO MASTER MODEL SPECIFICATION

⚠ CRITICAL LIMITATION — READ FIRST

The .std files themselves were **NEVER** uploaded to this project. Everything in Part D was extracted from **nineteen screen captures** of the STAAD.Pro interface. Node tables, member tables, load-case titles and reaction tables were read directly from those images and are marked **[CONFIRMED]**. Anything derived by back-calculation is marked **[RECONSTRUCTED]**. Anything not visible in a screenshot is marked **[NOT AVAILABLE — DO NOT INVENT]**.

A future Claude must ask the user to upload the actual .std files before making any claim about the model that is not listed below.

D.1 Model inventory

Model	File name shown in the title bar	Status
Underground box	Underground_Structure_WITH_LOADS_worked_example (4)	[CONFIRMED]
Sentry post	Sentry_Post_Framed_Seismic	[CONFIRMED]

Units: metres and kilonewtons throughout (node table headed “X m / Y m / Z m”, reactions “kN”, moments “kN-m”). [CONFIRMED] **Coordinate system:** STAAD global — **Y is vertical**, X and Z horizontal. [CONFIRMED]

D.2 SENTRY POST MODEL — Sentry_Post_Framed_Seismic

D.2.1 Nodes — [CONFIRMED, read from the geometry table]

Node	X (m)	Y (m)	Z (m)	Level
1	0.175	0.450	0.175	base, grid A-1
2	0.175	0.450	4.825	base, grid A-2
11	3.825	0.450	0.175	base, grid B-1
12	3.825	0.450	4.825	base, grid B-2
101	0.175	3.650	0.175	first floor, A-1
102	0.175	3.650	4.825	first floor, A-2
111	3.825	3.650	0.175	first floor, B-1
112	3.825	3.650	4.825	first floor, B-2
201	0.175	6.700	0.175	roof, A-1
202	0.175	6.700	4.825	roof, A-2
211	3.825	6.700	0.175	roof, B-1
212	3.825	6.700	4.825	roof, B-2
213	—	—	—	[NOT AVAILABLE] — row visible but values cut off by the scroll

Node numbering convention [RECONSTRUCTED]: base = 1/2/11/12, first floor = +100, roof = +200; the second digit pair encodes the grid (01 = A-1, 02 = A-2, 11 = B-1, 12 = B-2).

D.2.2 Members — [CONFIRMED, read from the beam table]

Beam	Node A	Node B	Property ref	Interpretation
1	1	101	1	ground-storey column A-1
2	2	102	1	ground-storey column A-2
3	11	111	1	ground-storey column B-1
4	12	112	1	ground-storey column B-2
5	101	201	1	first-storey column A-1
6	102	202	1	first-storey column A-2
7	111	211	1	first-storey column B-1
8	112	212	1	first-storey column B-2
9	101	111	2	first-floor beam, grid 1, spans X (B1 , 3650)
10	102	112	2	first-floor beam, grid 2, spans X (B1 , 3650)
11	101	102	2	first-floor beam, grid A, spans Z (B2 , 4650)
12	111	112	2	first-floor beam, grid B, spans Z (B2 , 4650)
13	201	211	2	roof beam, grid 1 (B1)
14	202	212	2	roof beam, grid 2 (B1)
15	201	202	2	roof beam, grid A (B2)
16	211	212	2	roof beam, grid B (B2) — [RECONSTRUCTED] , row not visible but required for a closed frame

Property references [RECONSTRUCTED from the drawings, consistent with the member layout]: - Property 1 = column, 350 × 350 rectangular - Property 2 = beam, 250 × 450 rectangular

Member releases / offsets: [NOT AVAILABLE] — no release or offset table was captured. The reaction pattern (all six components present at every base node, beam-end moments transferred to columns) is consistent with **no releases**, but this must be confirmed against the .std file.

Slab S1 is NOT modelled as plates. The slab load is applied to the beams as triangular/trapezoidal loading per the framing-plan schedule. [RECONSTRUCTED — no plate table appears in any sentry post screenshot, and the framing plan gives explicit peak beam intensities.]

D.2.3 Supports — [CONFIRMED]

Type: FIXED at all four base nodes (1, 2, 11, 12) at Y = +0.450. Evidence: the support-reaction table reports **all six components** (Fx, Fy, Fz, Mx, My, Mz) at every base node, and the plot symbols are the STAAD fixed-support hatched square.

Justification [CONFIRMED, from the design]: the columns sit on isolated footings on **in-situ rock**. Rotational restraint is real. On backfill it would not be defensible.

D.2.4 Load cases — [CONFIRMED, load-case titles read verbatim]

#	Title as it appears in STAAD
1	DL SELFWEIGHT SLAB INFILL PARAP... (truncated in the capture)
2	LL AS PER IS 875 PART 2
3	EQ+X IS 1893 VB 73.18 KN
4	EQ-X IS 1893 VB 73.18 KN
5	EQ+Z IS 1893 VB 73.18 KN
6	EQ-Z IS 1893 VB 73.18 KN

Applied load totals — [CONFIRMED, from the Statics Check Results table]:

L/C	ΣF _x	ΣF _y	ΣF _z	ΣM _x	ΣM _y
1 (DL)	0.000	−624.548	0.000	1561.369	0.000
2 (LL)	0.000	−76.376	0.000	190.941	0.000
3 (EQ+X)	73.180	0.000	0.000	0.000	182.950
4 (EQ−X)	−73.180	0.000	0.000	0.000	−182.950
5 (EQ+Z)	0.000	0.000	73.180	436.631	−146.360

Difference between Loads and Reactions = 0.000 in every case → **statics check passes**. [CONFIRMED]

Back-calculated load geometry — [RECONSTRUCTED]:

LC1 $M_x/F_y = 1561.369/624.548 = 2.500 \text{ m}$ → Z-centroid of dead load = mid-width ✓
LC3 $M_y/F_x = 182.950/73.180 = 2.500 \text{ m}$ → Z-centroid of the X seismic load ✓
LC5 $M_y/F_z = -146.360/73.180 = 2.000 \text{ m}$ → X-centroid of the Z seismic load ✓
LC5 $M_x/F_z = 436.631/73.180 = 5.967 \text{ m}$ → HEIGHT centroid above $Y = 0$

Solving the two-level distribution from the height centroid:

$Q_r \times 6.700 + Q_f \times 3.650 = 436.631$ and $Q_r + Q_f = 73.180$

→ $Q_{\text{roof}} = 55.57 \text{ kN}$ at +6.700

→ $Q_{\text{floor}} = 17.61 \text{ kN}$ at +3.650

Compare the hand $W h^2$ distribution: $Q_{\text{roof}} 59.51$, $Q_{\text{floor}} 13.67$.

→ THE MODEL'S VERTICAL DISTRIBUTION IS SLIGHTLY DIFFERENT FROM THE HAND CHECK.

Design used the hand distribution at the STAAD total (73.18 kN), which gives a

LARGER column moment at the ground storey. Conservative. See CONFLICT C9.

D.2.5 Support reactions, node 1 — [CONFIRMED, read from the reaction table]

L/C	$F_x \text{ kN}$	$F_y \text{ kN}$	$F_z \text{ kN}$	$M_x \text{ kNm}$	$M_y \text{ kNm}$
1 DL SELFWEIGHT	3.519	156.137	6.237	6.573	0.000
2 LL	0.588	19.094	1.209	1.272	−0.000
3 EQ+X	−18.295	−36.127	−0.000	−0.000	0.000
4 EQ−X	18.295	36.127	0.000	0.000	−0.000
5 EQ+Z	−0.000	−27.891	−18.295	−36.078	0.000
6 EQ−Z	0.000	27.891	18.295	36.078	−0.000
101 1.5DL+1.5LL	6.160	262.847	11.168	11.767	0.000
102	−17.026	166.926	8.935	9.413	−0.000
103	26.882	253.630	8.935	9.413	−0.000
104	4.928	176.808	−13.019	−33.881	0.000
105	4.928	243.747	30.889	52.707	0.000
106 1.5DL+1.5EQ+X	−22.165	180.016	9.355	9.859	0.000
107 1.5DL+1.5EQ−X	32.720	288.396	9.355	9.859	0.000

Two key confirmations from this table: 1. $F_x = 18.295 \text{ kN per column} \times 4 = 73.18 \text{ kN}$ — the portal-method distribution used in design is exactly what the model produces. [CONFIRMED] 2. $F_y = \pm 36.127 \text{ kN (EQ+X)}$ and $\pm 27.891 \text{ kN (EQ+Z)}$ — these ARE the seismic axial couples used in the column design. They were read from the model, not calculated by hand. [CONFIRMED]

D.2.6 Analysis and design commands — [NOT AVAILABLE — DO NOT INVENT]

No screenshot shows the STAAD editor, the PERFORM ANALYSIS line, P-Delta settings, DESIGN BEAM/DESIGN COLUMN blocks, or any design parameter (FYMAIN, FC, RATIO, TRACK, CLEAR, etc.). A future Claude must not state what these are. Ask for the .std file.

What the design work in this project actually used instead: hand calculations to IS 456 and IS 13920 (Part B.8), taking only the forces from the model. No STAAD concrete-design output was relied upon.

D.3 UNDERGROUND MODEL — Underground_Structure_WITH_LOADS_worked_example (4)

D.3.1 Model form — [CONFIRMED from the plots]

A plate + beam mid-surface model of the whole box: mat, four perimeter walls, internal walls and the pressure slab, meshed on a regular grid. The plots show a dense orthogonal mesh over the full 22 m × 6.2 m footprint with support symbols at every mat node.

Beam table shows 1139 members. [CONFIRMED] Plate table: [NOT AVAILABLE] — the plate list was never displayed in any capture.

D.3.2 Nodes — partial, [CONFIRMED for the rows visible]

Node	X (m)	Y (m)	Z (m)
1	0.300	0.300	0.3
2	0.300	0.300	0.8
3	0.300	0.300	1.3
4	0.300	0.300	2.0
5	0.300	0.300	2.7
6	0.300	0.300	3.3
7	0.300	0.300	4.0
8	0.300	0.300	4.6
9	0.300	0.300	5.2
10	0.300	0.300	5.9
11	0.825	0.300	0.3
12	0.825	0.300	0.8
13	0.825	0.300	1.3
14	0.825	0.300	2.0
15	0.825	0.300	2.7

Interpretation [RECONSTRUCTED — important for anyone rebuilding this model]:

The model uses MID-SURFACE geometry, and the model origin Y = 0 is at the UNDERSIDE OF THE MAT, i.e. real level (-)6.700.
Y = 0.300 = the mid-plane of the 600 mat ✓
Z = 0.300 and Z = 5.900 = the mid-planes of the 600 perimeter walls (real Y-plan 0-600 → mid 300 ; 5600-6200 → mid 5900) ✓
X = 0.300 = the mid-plane of the 600 west end wall ✓
Mesh pitch in Z ≈ 0.5 to 0.7 m ; in X the second line is at 0.825

This means the model's Y-axis is offset from the project level datum by 6.700 m. Project level = model Y - 6.700. A future Claude must apply this offset when comparing model output to the drawings.

All other nodes: [NOT AVAILABLE] — only rows 1-15 were captured.

D.3.3 Supports — [RECONSTRUCTED]

The plots show support symbols at every node of the mat, and the design basis (Part A.6) specifies a modulus of subgrade reaction of 100 000-500 000 kN/m³. This is consistent with elastic foundation (spring) supports, vertical only, on every mat node, generated from k_s.

The actual STAAD command (ELASTIC MAT, PLATE MAT, or explicit SPRING entries) is [NOT AVAILABLE].

⚠ **MODELLING WARNING THAT MUST NOT BE LOST:** spring supports in STAAD take tension unless explicitly declared compression-only. In the model the mat therefore never lifts, and the flotation check of Part B.3 CANNOT be performed in STAAD. It is a hand calculation on real dimensions and real weights.

D.3.4 Load cases — [CONFIRMED, titles read verbatim]

#	Title as it appears in STAAD	Value
1	DL1 SELF WEIGHT (SELFWEIGHT Y -1)	auto
2	DL2 EARTH COVER ON ROOF 40.65 KN/M2	40.65 kPa
3	DL3 SIDL SERVICES AND FINISHES	2.0 kPa
4	LL1 LIVE LOAD ON INTERNAL FLOOR 5.0 KN/...	5.0 kPa
5	DL4 STAIRCASE ON THE TWO SHAFT WAL...	2.947 kPa
6	EP1 EARTH + WATER ON EXTERNAL WAL...	33.9 → 83.2 kPa
7	HY1 HYDROSTATIC UPLIFT ON MAT 46.1 KN...	46.11 kPa
8	BL1 BLAST ON ROOF 383 KN/M2	383 kPa
9	BL2 BLAST ON EXTERNAL WALLS 383 KN/...	383 kPa
10	LL2 CONSTRUCTION SURCHARGE 20.0 KN	20.0 kPa

Load combinations — [CONFIRMED, titles read verbatim]

#	Title
101	ULS STATIC 1.5(DL+LL+EARTH+WATE...
102	ULS UPLIFT 0.9DL + 1.5 HYDROSTATIC
103	BLAST 1.0(DL + EARTH + WATER + BLAST...
104	SLS 1.0(DL+LL+EARTH+WATER) FOR CR...
105	CONSTRUCTION 1.5(DL + EARTH + WATE...

A Load Envelopes entry exists in the tree but its contents were [NOT AVAILABLE].

D.3.5 Underground model results — [NOT AVAILABLE — DO NOT INVENT]

No post-processing screenshot for the underground model was provided: no plate moments, no reactions, no deflections, no design ratios, no warnings.

Every underground design value in Part B was produced by hand calculation, not by STAAD output. The model provides the demand distribution and the visual check that loads are applied to the correct faces.

D.4 What a future Claude must obtain before extending the STAAD work

1. The two .std files themselves — everything above is screenshot-derived.
2. The plate table for the underground model (thicknesses, connectivity, material).
3. The support definition for the underground model, and whether the springs are compression-only.
4. The analysis and design command blocks for both models.
5. The STAAD seismic-weight summary for the sentry post, to close Conflict C9.
6. The underground model's post-processing output — plate M_x/M_y, support reactions, deflections.

PART E — AUTOCAD / DXF MASTER SPECIFICATION

E.1 Two distinct DXF sets — do not confuse them

Set	Origin	Revision	Purpose
Input set (10 files)	Supplied by the user	Rev F	Architectural / civil geometry — the PRIMARY GEOMETRY SOURCE
Output set (8 files, S-01 to S-08)	Generated in this project	Phase 2 Rev A	Structural drawings with reinforcement

E.2 Input DXF set — Rev F — [CONFIRMED, all parsed]

File	Content	Key extracted geometry
1_Underground_Level_Plan.dxf	GA plan — PRIMARY	Box 0–21600 × 0–6200; internal 600–21000 × 600–5600; partitions; internal walls; ESC circles r=700/r=950; stair shaft; blast door swings r=1200
1_Staircase_Section.dxf	Section A-A, main stair	Landings 200 thk; waist 200 (from headroom 2533); flights; store under L1
2_Underground/Side_Section_with_Stairs.dxf	Section X-X	Contains the “REV F” marker and the covered-stairwell note
2_Ground_Plan_Headhouse_Berm.dxf	Ground plan	Headhouse 13400–18200 × 200–6000; stairwell 9050–15850 × 5750–7750; berm; grating
3_Headhouse_Section_Cutaway.dxf	Section B-B at Z = 2800	Shaft, Bay 6 and Bay 8 either side
5_Entry_Headhouse_Stair_Section.dxf	Section C-C, Rev F	12R @ 166.667/300; raking roof 250; platform 250; entry door 1000×2100
5_Front_Elevation.dxf	Front elevation	Sentry post levels +0.450 / +3.650 / +6.700 / +7.000
3_Sentry_Post_Ground_Floor_Plan.dxf	Sentry GF	4000 × 5000 external; 200 RC infill; D1 900
4_Sentry_Post_First_Floor_Plan.dxf	Sentry FF	Armoured vision panels 1200 wide
6_Sentry_Post_Framing_Plan.dxf	Sentry framing — PRIMARY for the frame	Grid A-B 3650, 1-2 4650; C1 350×350; B1/B2 250×450; S1 150; PB 250×400; the full load schedule

Input set drawing conventions [CONFIRMED]: - Units: millimetres - Origin: SW external corner of the box at (0, 0) - Layers observed: WALLS, STAIRS, OPENINGS, GROUND, CENTRE, DIM, TEXT, HATCH - Entities: LINE, POLYLINE + VERTEX + SEQEND, CIRCLE, ARC, TEXT - Sections use local coordinate systems — e.g. in Section A-A the local x maps to the project plan Y (0 = box south external face)

E.3 Output DXF set — S-01 to S-08 — [CONFIRMED, all generated in this project]

E.3.1 Sheet standard — applies to every output sheet

SHEET SIZE	A1, 841 × 594 mm, DRAWN IN PAPER MILLIMETRES, PLOT 1:1
DXF FORMAT	AutoCAD R12 ASCII (maximum compatibility; opens in AutoCAD, DraftSight, BricsCAD, LibreCAD, QCAD without translation)
BORDER	outer 0,0–841,594 ; inner 10,10–831,584
TITLE BLOCK	180 × 62 mm, bottom right, at x0 = 651, y0 = 10
VIEW SCALES	each view carries its own scale note; views are mapped from model millimetres to paper millimetres by paper = origin + model/scale
TEXT HEIGHTS	3.4 view titles 2.4 panel headings 2.0 scale notes 1.6–1.7 general 1.55–1.62 panel body 1.4–1.5 small annotation
VALIDATION	all eight files verified: SECTION/ENDSEC balanced, EOF present, no undeclared layers, all geometry within 0..841 × 0..594

E.3.2 Layer table — 22 layers, identical in every output file [CONFIRMED]

Layer	Colour	Purpose
0	7	Default, unused
CONC	7	Concrete outlines
CONC - HIDDEN	8	Concrete beyond / hidden
REINF - MAIN	1 (red)	Main reinforcement
REINF - DIST	3 (green)	Distribution reinforcement
REINF - LINK	6 (magenta)	Links, stirrups, torsion mats
REINF - SEC	4 (cyan)	Secondary / trimmers / door frames
DIM	4	Dimensions and balloons
TEXT	7	General annotation
TEXT - TITLE	7	View titles, headings
HATCH	8	Section hatch
GRID	5	Grid lines
SOIL	8	Soil, rock, berm, cover layers
CENTRELINE	6	Centrelines, void diagonals
TITLEBLOCK	7	Border and title block
WATERPROOF	30 (orange)	Membranes, waterstops, GWT
STEELWORK	2 (yellow)	EMP straps, steel items
LEVELS	4	Level markers
NOTES	7	Panel body text
TABLE	7	Table rules and load diagrams
SERVICES	2	Sumps, tanks, ducts, drainage
BLAST	1 (red)	Blast arrows, findings, warnings

All layers: 70 = 0 (on, thawed, unlocked), linetype CONTINUOUS. Dashed lines are drawn as **explicit line segments**, not by linetype, for maximum compatibility.

E.3.3 Output sheet register

Sheet	File	Title	Key views
S-01	01_Shear_Wall_Structural_Drawing.dxf	Shear walls / retaining walls	V1 typical perimeter wall section 1:25 with 9-item balloon key; V2 W6/W7 400 section; V3 wall elevation 1:50; V4 haunch 1:10; V5 construction joint 1:5; V6 lateral load diagram; V7 wall layout key plan 1:200; wall schedule; BBS; 3 panels incl. Finding F1
S-02	02_Mat_Foundation_Structural_Drawing.dxf	Mat foundation	V1 mat plan 1:60; V2 section A-A mat/wall junction 1:20; V3 mat edge/drainage 1:10; V4 section B-B sump pit 1:25; bearing table; staged flotation table ; mandatory mitigation; BBS; further checks
S-03	03_Roof_Slab_Structural_Drawing.dxf	Roof (pressure) slab	V1 roof plan 1:60; V2 section C-C 1:40; V3 1400 dia collar 1:30; V4 section D-D stair-void free edge 1:25; roof loading table; thickness study; BBS; Finding F2 panel
S-04	04_Main_Staircase_Structural_Drawing.dxf	Main staircase, Bay 7	V1 stair plan 1:35; V2 section A-A longitudinal 1:30; V3 flight/landing junction 1:20; V4 landing section 1:30; NBC geometry check; BBS
S-05	05_Approach_Stairwell_Structural_Drawing.dxf	Approach stairwell + headhouse	V1 plan 1:65; V2 section C-C 1:50; V3 section E-E transverse 1:40; V4 raft/headhouse movement joint 1:20; element schedule; BBS; C2 + C10 headhouse panel
S-06	06_Underground_Plan_Services_Sump_BlastValves.dxf	Services and drainage	V1 underground plan 1:60 with 5 blast valves marked ; V2 site drainage 1:300; V3 sump pit section 1:30; V4 septic tank + soak pit 1:40; V5 filter train schematic; blast valve schedule; drainage flows
S-07	07_Entry_Stairwell_Flight_Reinforcement.dxf	Entry stairwell flight	V1 longitudinal section 1:30; V2 plan on bottom steel 1:30; V3 section A-A 1:20; V4 the opening corner 1:20 ; V5 foot of flight 1:20; NBC geometry check; BBS
S-08	08_Sentry_Post_Slab_Reinforcement.dxf	Sentry post slab S1	V1 bottom steel plan 1:30; V2 top + torsion plan 1:30; V3 section A-A slab/edge beam 1:10; V4 corner torsion mat 1:20; design summary; BBS; loads; “why Table 26 not Table 27” panel

Not yet drawn: sentry post beams, columns and footings. A future sheet **S-09** would carry B1 4-T16/2-T16, B2 3-T20/2-T20, C1 8-T16 with T10 confining hoops @ 85, and F1 1500²×600 with T12 @ 150 B/W.

E.4 The DXF generation toolchain — CRITICAL FOR REGENERATION

The output DXFs were not hand-drafted. They are produced by a dependency-free Python toolchain built inside this project. ezdxf is NOT available in the environment and the network is disabled — the writer is hand-rolled.

File	Purpose
dxflib.py	AutoCAD R12 ASCII DXF writer. Class Dxf with methods: line, circle, arc, text, rect, dline (dashed), cline, hatch45, soilhatch, tick, dimh, dimv, leader, level, secmark, table, notes, titleblock, save. Holds the 22-layer table.
proj.py	All reconciled project constants — PROJ, M1, S, L (levels), T (thicknesses), BOX, INT, PART_X, DOOR_Y, IW, SHAFT, VOID, PAD, FLTA, FLTB, BAY8, ESC1, ESC2, HH, HH1, ASW, ASWI, ASW_TOP, ASW_FLT, ASW_PLAT, plus text blocks MAT_NOTES, LOADS, CODES, CONFLICTS, MOD1, MOD2 and helpers sheet, vw, vtitle, panel, balloon, keylist.
d01_wall.py ... d08_sentryslab.py	One generator per sheet
parse.py	DXF reader (read, ents)
geo.py, txt.py	Geometry / text extractors used to mine the input DXFs
render.py	Matplotlib visual-QA renderer, DXF → PNG
render_pdf.py	DXF → true-size A1 vector PDF , 1:1
validate.py	Structural validation of every output DXF
calc.py, colpm.py	Numeric verification: flexure solver, τ_c Table 19 interpolation, column P-M interaction

Key mapping helper — `vw(sc, ox, oy)` returns a function $P(\text{model_x}, \text{model_y}) \rightarrow (\text{paper_x}, \text{paper_y})$ where $\text{paper} = \text{origin} + \text{model}/\text{scale}$. Every view uses one.

A future Claude editing a drawing should edit the generator and re-run it, not edit the DXF. The DXF is a build artefact.

PART F — REINFORCEMENT MASTER REGISTER

Notation: T16 @ 150 EF EW = 16 mm Fe500D deformed bars at 150 mm centres, **Each Face**, **Each Way**. 4L = four-legged links.

F.1 Underground box

Element	Size	Cover	d (mm)	Main reinforcement	Links	A _{st} prov	x _u /d	Util.
W1 south perimeter	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W2 north perimeter	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W3 west end	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W4 east end	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W5 Bay 5/6	200	40	154	T12 @ 150 EF EW	—	754/face	—	nominal
W6 Bay 6/7 (M1)	400	50/40	342	T20 @ 150 EF EW	T12 4L @ 200	2094/face	0.211	69 %
W7 Bay 7/8 (M1)	400	50/40	342	T20 @ 150 EF EW	T12 4L @ 200	2094/face	0.211	69 %
W8 partitions x4	110	25	—	A252 mesh both faces	—	252	—	non-structural
MAT	600	75/50	517	T16 @ 150 EF EW	T12 @ 250 x 250 grid	1340/face	0.089	84 %
ROOF midspan	900	75	812.5	T25 @ 150 EF EW	T12 4L @ 250 end 1500 / 2L @ 300 mid	3272/face	0.139	51 %
ROOF cantilever pad	900	75	812.5	T25 @ 150 top continuous	T12 4L @ 250 throughout	3272	0.139	56 %
Sump pit walls/base	300/400	50	244	T16 @ 150 EF EW	—	1340	—	nominal

Additional bars — underground box

Location	Requirement
Wall starters	T16 @ 150 EF, 900 mm horizontal leg into the mat , lap 800 (50φ) above a 150 kicker
Haunches, all wall-roof and wall-mat junctions	500 × 500 , diagonal T20 @ 150
Mat edge, all free edges	T16 @ 150 U-bars closing both curtains
Sump pit opening	4-T20 trimmers each face, each side , L _d 800 beyond
Blast door jambs (W6, W7)	4-T20 each jamb, each face , L _d 800 beyond
Blast door header	400 × 1100: 4-T20 top + 4-T20 bottom, T12 4L @ 150
ESC 1 / ESC 2 collars	Collar 250 RC; 5-T25 each side, each face, each direction , L _d 1000 beyond; T16 @ 150 hoops ×3 layers + T16 @ 150 radials ; slab thickened 900→1200 over a 600 annulus
Stair void free edge	Thickened 900→1200 over 600; 6-T25 top + 6-T25 bottom, T12 closed @ 150
Stair void side bands	6-T25 each face, top and bottom, in a 900 band over W6 and W7
Stair void re-entrant corners (×2)	4-T25 each face at 45°, 2000 long each way , anchored L _d beyond
Band beneath headhouse wall HW3	4-T25 extra top and bottom in a 1200 band , lapped 1250
Construction joints (~6 m)	Reinforcement fully continuous ; two waterstops + welded Cu/galv EMP strap

F.2 Stairs

Element	Size	Cover	d	Main	Distribution	Top steel
Main stair flight waist	200	30	164	T12 @ 150 (754)	T10 @ 200 (393)	T12 @ 150 for 800 into the flight, L _d 480
Main stair landings L1/L2/arrival	200	30	164	T12 @ 125 bottom (905)	T10 @ 200	T12 @ 125 for 900 from each support
Entry stairwell flight waist	250	30	214	T16 @ 200 (1005)	T10 @ 200	T16 @ 200 for 1200 into the flight, L _d 640
Entry stairwell top landing	250	30	214	T12 @ 200 EF EW (565)	—	—
Entry stairwell platform	250	30	214	T12 @ 200 EF EW	—	on fill
Entry stairwell side walls	250	50	194	T12 @ 200 EF EW	—	min steel governs
Entry stairwell raking roof	250	30	204	T12 @ 200 EF EW	—	min steel governs
Entry stairwell headwall	250	30	214	T12 @ 200 EF EW	—	—
Entry door lintel (1000 clear)	250 × 350	30	306	3-T12 top + 3-T12 bottom	T8 @ 150	—
Stepped raft	300	50	244	T12 @ 200 EF EW	—	on compacted fill
Opening-corner U-bar (top of entry flight)	—	—	—	T16 @ 200 U-bar , bars CROSSED and anchored L _d 640 into the opposite face		SP 34 Cl. 5.5

F.3 Headhouse

Element	Size	Cover	d	Main	Links
Roof	500	75	415	T20 @ 150 EF EW (2094)	T12 4L @ 175 in the end 1200 each side; @ 250 elsewhere
Walls HW1-HW4	400	50/40	342	T16 @ 150 EF EW (1340)	T12 4L @ 250 throughout
Door jambs, HW2	—	—	—	2-T20 each jamb, each face , L _d 800 beyond	—
Door head edge band	400 × 800	40	742	4-T20 top + 4-T20 bottom	T12 4L @ 150, over the opening + 600 each side
Wall-roof haunch	300 × 300	—	—	diagonal T16 @ 150	—

F.4 Sentry post

Element	Size	Cover	d	Reinforcement
Slab S1 bottom	150	30	116/108	T8 @ 150 c/c BOTH WAYS (335 mm ² /m)
Slab S1 top, discontinuous edges	150	30	116	T8 @ 300, 400 mm (0.1 L) into the span, all four edges — Cl. D-1.6
Slab S1 corner torsion	—	—	—	T8 @ 200, FOUR LAYERS, 700 × 700 at all four corners — Cl. D-1.8
Beam B1 (3650)	250 × 450	30	404	4-T16 top at supports; 2-T16 bottom continuous
Beam B2 (4650)	250 × 450	30	404	3-T20 top at supports; 2-T20 bottom continuous
Beam hoops, both	—	—	—	T8 2-leg @ 100 over 2d = 810 from each face; @ 150 elsewhere ; first hoop ≤ 50 from the face
Column C1	350 × 350	40	d' 56	8-T16 (1608 mm ² , $\rho = 1.31\%$)
Column confining hoops	—	—	—	T10 hoops + 1 cross-tie each way @ 85 c/c over 500 mm from every joint face, top and bottom, AND through the joint
Column general ties	—	—	—	T8 @ 150 c/c elsewhere
Plinth beam PB	250 × 400	30	354	3-T12 top + 3-T12 bottom
Footing F1	1500 × 1500 × 600	50	542	T12 @ 150 both ways bottom (754 mm ² /m)

3-T20 was chosen for B2 over 5-T16 because five T16 bars require 256 mm in a 250 mm wide beam. That is exactly the check that becomes a site RFI if skipped.

PART G — CODES, STANDARDS AND REFERENCES

Code / reference	Used for	Specific clauses relied upon
IS 456:2000	All RC design	Cl. 6.2.3.1 (E_c); Cl. 13.4 (construction joints); Cl. 22.2(a) (effective span); Cl. 23.2.1 + Fig. 4 (deflection); Cl. 24.4, 24.5 + Fig. 7 (two-way slabs, loads to beams); Cl. 25.1.2 (short column); Cl. 25.4 (min eccentricity); Cl. 26.2.1, 26.2.1.1, 26.2.5.1 (L_d , laps); Cl. 26.3.3 (bar spacing); Cl. 26.4.2, 26.4.2.1, 26.4.2.2 + Table 16 (cover); Cl. 26.5.1.1, 26.5.1.2, 26.5.1.5, 26.5.1.6 (beam steel, links); Cl. 26.5.2.1, 26.5.2.2 (slab steel); Cl. 26.5.3.1, 26.5.3.2 (column steel, ties); Cl. 31.6, 31.6.1, 31.6.3.1 (punching); Cl. 32.2, 32.5(a)(b)(c) (walls); Cl. 33.1(b), 33.2 (stairs); Cl. 34.2.4.1 (footing shear); Cl. 36.4.2 (γ_m); Cl. 38.1, 38.1(c), 38.1(f) (limit state, stress block, $x_{u,max}$); Cl. 39.1, 39.3, 39.6 (columns, biaxial); Cl. 40.1, 40.2.1.1, 40.2.3, 40.4(a) (shear); Table 3, 5, 16, 18, 19, 20, 26, 27, 28; Annex D (D-1.1 to D-1.8, D-2); Annex G-1.1(b), G-1.1(c)
IS 875 (Part 1):1987	Dead loads, unit weights	Table 1
IS 875 (Part 2):1987	Imposed loads	Plant/storage 5.0 kPa; office/OP 3.0; accessible roof 1.5
IS 875 (Part 3):2015	Wind, sentry post only	Cl. 6.3, 6.3.4, 7.2, 7.2.1, 7.3.3.13, 7.4; Tables 1, 2, 4, 26; Annex A ($V_b = 39$ m/s)
IS 875 (Part 5):1987	Load combinations	with IS 456 Table 18
IS 1893 (Part 1):2016	Seismic	Cl. 6.3.1.2, 6.3.2.2 (combinations, directional); Cl. 6.4.2 + Fig. 2 (A_h , S_a/g); Cl. 7.2.3 (I), Cl. 7.2.6 (R), Cl. 7.3.1, 7.3.2 (seismic weight); Cl. 7.6.1, 7.6.2, 7.6.2(c), 7.6.3 (period, base shear, distribution); Cl. 7.11.1 (drift); Table 3 (Z)
IS 13920:2016	Ductile detailing, sentry post	Cl. 6.1.1, 6.1.2, 6.1.3 (beam geometry); Cl. 6.2.1(b), 6.2.2, 6.2.3, 6.2.4 (beam steel); Cl. 6.3.3, 6.3.4 , 6.3.5.1, 6.3.5.2 (capacity-design shear, hoops); Cl. 7.1, 7.2.1 (SCWB), 7.3, 7.4 (columns); Cl. 8.1, 8.2 (special confining reinforcement); Cl. 10.4 (boundary elements — checked, not triggered)
IS 3370 (Parts 1, 2):2021	Crack width, surface-zone steel	0.2 mm limit; 0.35 % surface-zone steel over a 250

IS 3370 (Parts 1, 2):2021	Crack width, surface-zone steel	0.2 mm limit, 0.33 % surface-zone steel over a 250 mm zone
IS 1786:2008	Fe500D reinforcement	D grade — guaranteed minimum elongation
IS 1904:1986	Presumptive bearing capacity	Table 1 (hard rock, 3240 kPa)
IS 2950 (Part 1):1981	Raft foundations	General provisions
IS 12070	Rock foundations	Cl. 6 (settlement on sound rock)
IS 2470 (Part 1):1985	Septic tank	Cl. 6.2 (24 h detention), Cl. 6.3 (30 L/person/yr sludge), Cl. 6.5 (L:B 2–4), Cl. 6.6 (min width 750, min depth 1.0 m), Cl. 6.9 (vent); Table 1 (≤10 users: 1.5 × 0.75 × 1.0)
IS 2470 (Part 2):1985	Soak pit / dispersion	Cl. 4 (percolation test — mandatory), Cl. 5 (dispersion trench)
IS 4991:1968	Blast loading rules and dynamic material strengths ONLY	Cl. 1.1 (EXPLICITLY EXCLUDES NUCLEAR — quoted in every deliverable) ; Cl. 6.2.1 (reflected pressure, recessing); Cl. 7.2 + Table 3 (buried roof/walls, K_a); Cl. 7.4 (drag on bermed faces); Cl. 10.3.1 (dynamic f_{ck} , f_y); Cl. 10.3.1.1 (NO dynamic increase on shear; +25 % bond) ; Cl. 10.3.3 ($DLF = \mu/(\mu-0.5)$); Cl. 11.1 (no wind/EQ with blast); Cl. 11.2 (no live load on the roof at blast); Fig. 6 (SDOF — Phase 3)
SP 16:1980	Design aids	Column interaction charts (cross-checked; the P-M values in B.8.6 were computed from first principles, not read off)
SP 34:1987	Detailing	Shape codes; Cl. 5.5 (opening-corner detailing)
NBC 2016 Part 4	Fire and life safety	Stair geometry, guarding 1100, means of escape
BS 8666	Bar shape codes	Used alongside SP 34 in all BBS
UFC 3-340-02	US criteria, cited AS US criteria	§4-27 (opening trimmers); §4-30 (direct shear)
UFC 3-350-04AN	US criteria	Referenced in the Phase 1 report
MIL-STD-188-125-1	EMP	§5.4 (access), §5.5 (waveguide below cutoff), §5.7.2.1 (power PCI), §5.7.4.1 (fibre), §5.7.6 (RF); 80 dB, 10 kHz–1 GHz
IEEE Std 299	EMP verification	Shielding effectiveness survey
IEEE 142 / IS 3043	Earthing	≤ 5 Ω target
FEMA 453	Shelter ventilation	0.25 cfm/ft ² ; carbon adsorber performance
EN 1822	HEPA classification	H14, ≥ 99.995 % at MPPS
Glasstone & Dolan	Weapons effects	Unclassified, public — blast and radiation data only
Biggs, <i>Structural Dynamics</i>	SDOF method	Referenced for the Phase 3 support-rotation check

No clause number in this register was invented. Where a clause could not be confirmed from the material available it is not listed. If a future Claude needs a clause not in this table, it must be verified against the code, not assumed.

PART H — REVISION HISTORY

H.1 Architectural / Phase 1 history (pre-dating this work)

Rev	Change	Previous	New	Reason	Status
A-B	Initial layout development	—	—	—	SUPERSEDED
C	Box lengthened to give ESC 2 the 750 mm clearance	shorter box	21 600 external	Escape-shaft opening edge was < 750 mm from a wall face	SUPERSEDED by M1
—	Cover reduced	4 000 mm	2 000 mm	Prompt radiation needs 2 m; blast needs none; fallout needs 1 m. Saved 2 m excavation, 2 m shaft, one stair flight, 2 m headroom	CURRENT
D	Design report issued	—	—	Basis of design, loads, materials, geotech, calculations	CURRENT for loads/materials; SUPERSEDED for the entrance and the headhouse roof
D reg. item 11	Headhouse resized	3 000 × 4 000	4 000 × 5 000, no roof cover, +0.900	—	CURRENT — but O.12 was never recomputed → Conflict C2
E-F	Approach replaced	dog-leg open ramp	covered entry stairwell , 12R @ 166.667/300, entered at grade	Open cut had no gravity outfall; a 2 m pit collects dense CBRN agents; lidding at grade gave zero headroom	CURRENT — Conflict C1
F	Drawing set issued	—	—	The ten uploaded DXFs	CURRENT for all geometry

H.2 Phase 2 structural history (this project)

#	Change	Previous	New	Reason	Affected files	Status
M1	Walls W6/W7 thickened; box lengthened	200 thk; box 21 600; shaft 15000-17800; Bay 8 18000-21000; ESC 2 at 19500	400 thk; box 22 000; shaft 15200-18000; Bay 8 18400-21400; ESC 2 at 19900	Finding F1 — 200 mm cannot carry 383 kPa ($M_{u,lim}$ 138 < 245 demand). 400 cannot come from Bay 8 (750 clearance) or the shaft ($2800 = 2 \times 1200 + 200$)	ALL sheets; headhouse and stairwell shift +200	CURRENT — awaiting client approval
F2	Stair-void cantilever pad designed	not designed	T25 @ 150 top (56 %) + T12 4L @ 250 throughout the pad + edge thickening + corner trimmers	$M(\text{root})$ 758.6 > midspan 700.2	S-03	CURRENT
C6	Mat one-way shear checked	flexure only	T12 @ 250 × 250 link grid	IS 4991 Cl. 10.3.1.1 forbids dynamic increase on shear	S-02	CURRENT
C2	Headhouse roof recomputed	3.0 m span, 1.0 m cover	4.0 m span, no cover, w = 396.5 kPa	Report 0.12 stale vs Rev F	S-05	CURRENT
C2	Headhouse roof shear links added	none	T12 4L @ 175 end 1200 / @ 250	τ_v 1.514 > τ_c 0.502	S-05	CURRENT
C10	Headhouse wall loading raised	113 kPa drag (Cl. 7.4)	383 kPa full envelope	Berm transmits pressure; K_a unverifiable. Cost = one link cage	S-05	CURRENT
C10	Headhouse wall links added	none	T12 4L @ 250 all four walls	consequence of the above	S-05	CURRENT
C9	Sentry post base shear	hand calc 59.3 kN (W = 592.7)	STAAD 73.18 kN — used for all design	Model 23 % heavier; reactions confirm $F_x = 18.295/\text{column}$	Part B.8	CURRENT — gap to be closed in Phase 3
C9	Beam B1 top steel	3-T16 (at 59.3 kN)	4-T16	consequence of C9	Part B.8.4	CURRENT
C9	Beam B2 top/bottom	4-T16 / 2-T16	3-T20 / 2-T20	consequence of C9; 5-T16 will not fit a 250 wide beam	Part B.8.5	CURRENT
C9	Column biaxial ratio	0.718	0.819	consequence of C9	Part B.8.6	CURRENT
ERR-1	Yield-line coefficient corrected	324.4 kNm/m (used the simply-supported coefficient 24)	162.2 kNm/m (fixed-4-edges coefficient 48)	My own arithmetic error. Validation: as $b \rightarrow \infty$, $48/3 = 16 = w \cdot L^2/16$ ✓	Part B.7.1, S-05	CURRENT. No reinforcement change — the one-way value 396.5 was adopted, so nothing depended on it
Sump relocated	outside the wall, below mat level	inside, Bay 5, 1500³, invert (–)7.600	Unprotected envelope breach below the water table, impossible to inspect	S-02, S-06	CURRENT	
Wall links	T12 @ 250	T12 @ 200	Recalculated V_{us}	S-01	CURRENT	

H.3 Conflict register — full

#	Conflict	Resolution	Status
C1	Drawings Rev F , report Rev D . Report describes a dog-leg open ramp; Rev F is a covered entry stairwell	DXF governs . Part 6 of the calculation report designs the Rev F arrangement	RESOLVED
C2	Report O.12 designs the headhouse roof for 3.0 m span + 1.0 m cover; Rev F is 4.0 m + no cover	Recomputed . T20 @ 150 survives at 90 %; shear links new and mandatory	RESOLVED
C3	Report G.2 / M-9 call the stair-shaft walls 600 thk; the Rev F plan shows 200	DXF governs → triggers Finding F1 → Modification M1	RESOLVED
C4	Blast door leaf: plan opening 1200, Section A-A shows 900	Adopted 1200 × 2100 — agrees with report F.5 (1564 kN leaf force)	RESOLVED
C5	Plan dim “7300 covered stairwell external” vs 6800 in the plotted geometry and Section C-C	Adopted 6800	RESOLVED
C6	Report O.9 checks the mat in flexure only	One-way shear checked → link grid required	RESOLVED
C7	Report O.14 uses $W = 505 \text{ kN [A]}$ for the sentry post	Computed as 592.7 kN from the drawn sizes → 59.3 kN. Superseded by C9	SUPERSEDED
C8	Report adopts sentry post height 6850 mm; the front elevation shows roof +6.700, parapet +7.000	Drawn levels adopted ; $h = 6.250 \text{ m}$ from the base at +0.450	RESOLVED
C9	STAAD states $V_b = 73.18 \text{ kN}$; hand check gives 59.3 kN	Design uses the higher STAAD value . Print the STAAD seismic-weight summary in Phase 3	OPEN — documentation gap, not a safety gap
C10	Report designs headhouse walls for 113 kPa drag only	Raised to the full 383 kPa envelope . Cost: T12 4L @ 250 in all four walls	RESOLVED
—	Report J.7 places the service-entry plate “between Bay 5 and Bay 8” — those bays are not adjacent	Superseded . Single plate placed in the north perimeter wall (W2) at Bay 5, $X \approx 11\,800$	RESOLVED

PART I — PROJECT FILE MANIFEST

I.1 CURRENT FILES — input (user-supplied)

File	Type	Rev	Purpose	Status
1_Underground_Level_Plan.dxf	DXF	F	Primary geometry source	CURRENT
1_Staircase_Section.dxf	DXF	F	Section A-A, main stair	CURRENT
2_Side_Section_with_Stairs.dxf	DXF	F	Section X-X; carries the Rev F marker	CURRENT
2_Ground_Plan_Headhouse_Berm.dxf	DXF	F	Headhouse + stairwell plan, berm	CURRENT
3_Headhouse_Section_Cutaway.dxf	DXF	F	Section B-B	CURRENT
5_Entry_Headhouse_Stair_Section.dxf	DXF	F	Section C-C, the Rev F entry stairwell	CURRENT
5_Front_Elevation.dxf	DXF	F	Sentry post levels	CURRENT
3_Sentry_Post_Ground_Floor_Plan.dxf	DXF	F	Sentry GF	CURRENT
4_Sentry_Post_First_Floor_Plan.dxf	DXF	F	Sentry FF	CURRENT
6_Sentry_Post_Framing_Plan.dxf	DXF	F	Primary source for the sentry frame + load schedule	CURRENT
Phase1_Design_Report_RevD.md	Markdown, 126 KB	D	Primary source for loads, materials, geotech	CURRENT for loads/materials; SUPERSEDED for the entrance (C1) and headhouse roof (C2)
SK02_Underground_Plan.png	PNG	—	Coloured GA plan, presentation graphic	CURRENT
19 STAAD screen captures	PNG	—	The only evidence of the STAAD models	CURRENT

I.2 CURRENT FILES — output (generated in this project)

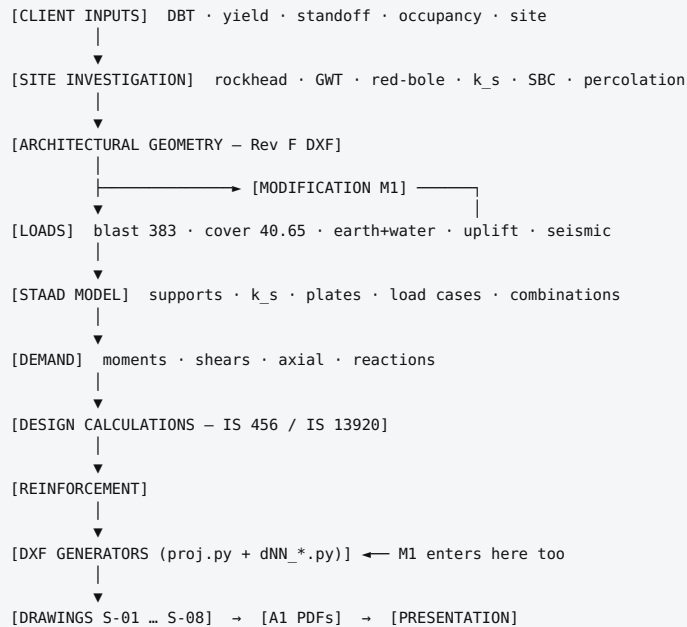
File	Type	Purpose
01_Shear_Wall_Structural_Drawing.dxf / .pdf	DXF + A1 PDF	S-01
02_Mat_Foundation_Structural_Drawing.dxf / .pdf	DXF + A1 PDF	S-02
03_Roof_Slab_Structural_Drawing.dxf / .pdf	DXF + A1 PDF	S-03
04_Main_Staircase_Structural_Drawing.dxf / .pdf	DXF + A1 PDF	S-04
05_Approach_Stairwell_Structural_Drawing.dxf / .pdf	DXF + A1 PDF	S-05
06_Underground_Plan_Services_Sump_BlastValves.dxf / .pdf	DXF + A1 PDF	S-06
07_Entry_Stairwell_Flight_Reinforcement.dxf / .pdf	DXF + A1 PDF	S-07
08_Sentry_Post_Slab_Reinforcement.dxf / .pdf	DXF + A1 PDF	S-08
00_Drawing_Set_S01_to_S08.pdf	PDF, 8 pages A1	Combined drawing set
Structural_Design_Calculations_IS456.md / .pdf	17 pp	Full calculation report with clause citations
Headhouse_Walls_and_Slab_Design.md / .pdf	8 pp	Dedicated headhouse design + the ERR-1 correction
Load_Calculations_Explained.md / .pdf	7 pp	Plain-language loads + STAAD modelling basis
Services_Drainage_EMP_BlastValves.md / .pdf	7 pp	Drainage, EMP cage, blast valves, ventilation
Consolidated_Member_Schedule.md / .pdf	5 pp	Every member, one table
Presentation_Speaking_Notes.md / .pdf	7 pp	90-minute run sheet + anticipated questions
Phase1_Design_Report_RevD.pdf	33 pp	The Rev D report, rendered
Python toolchain (18 files)	.py	DXF generators + verification scripts — see E.4

I.3 SUPERSEDED / ARCHIVED

Item	Superseded by	Note
Box length 21 600	22 000 (M1)	All pre-M1 dimensions east of X 14 800
W6/W7 at 200 thk	400 thk (M1)	Finding F1
ESC 2 at X 19 500	X 19 900	M1
Report Rev D approach ramp	Rev F covered stairwell	C1
Report O.12 headhouse roof	Recomputed, B.7.1	C2
Sentry V _b 50.5 kN (report) and 59.3 kN (hand)	73.18 kN (STAAD)	C7, C9
Yield line 324.4 kNm/m	162.2 kNm/m	ERR-1
Sump outside the wall	Inside, Bay 5	—
Wall links T12 @ 250	T12 @ 200	—
B1 3-T16 / B2 4-T16	B1 4-T16 / B2 3-T20	C9

⚠ **There is no file in the uploaded set that reflects Modification M1.** Every input DXF is at the pre-M1 geometry (box 21 600). **The M1 geometry exists only in the output drawings S-01 to S-08 and in this document.** A future Claude must not treat the input DXFs as current for anything east of X = 14 800.

PART J — DEPENDENCY MAP



J.1 Change-propagation matrix — WHAT MUST BE UPDATED IF ...

If this changes	Then these MUST be revisited
Any plan geometry	proj.py constants → all 8 DXF generators → all drawings → STAAD node/plate geometry → every affected calculation → member schedule
The 5 000 mm internal width	Roof $M_p \propto L_n^2$ → roof steel, roof links, thickness study, wall axial load, mat, flotation, quantities. The most expensive single change in the project.
Box length only	Nothing structural — the roof spans one-way across the width. Only quantities, uplift total, and drawing dimensions. This is why M1 is cheap.
Any thickness	$d \rightarrow A_{st,req} \rightarrow M_u \rightarrow$ utilisation → $x/d \rightarrow$ shear → links → cover check → BBS → DXF → self weight → seismic weight → flotation
Blast p_{so}, t_d, or μ	DLF → 383 kPa → every element of the box + headhouse roof + blast doors + blast valves + t_d/T quasi-static justification
Groundwater level	Earth+water gradient → wall design → uplift → flotation FoS at all 4 stages → dewatering strategy → waterproofing class → sump duty
k_s	Mat moments (sensitive) → mat steel → link grid. Run both bounds.
SBC	Bearing check only (currently 1.8 %/12.5 % — enormous margin)
Material grade	$E_c \rightarrow$ period $T \rightarrow t_d/T$; $f_{ck,dyn} \rightarrow$ all $A_{st,req}$; τ_c Table 19 → all links ; $\tau_{bd} \rightarrow$ all L_d and laps → all BBS cut lengths
Cover	$d \rightarrow$ every flexural result; also EMP spacing and IS 3370 surface-zone steel
Sentry post V_b (C9)	Column shears → beam moments → B1/B2 steel → capacity-design shear → hoop spacing → SCWB check → column biaxial → footing moment
R factor (3.0 vs 5.0)	$V_b \times 0.6 \rightarrow$ everything in the row above; plus the infill would need out-of-plane restraint
Support conditions in STAAD	Demand distribution → all design; and note flotation can never be taken from the model
Reinforcement	$M_{u,prov} \rightarrow$ utilisation → $x/d \rightarrow p_t \rightarrow \tau_c \rightarrow$ links → capacity-design shear (IS 13920) → SCWB → BBS → DXF
Headhouse geometry	Roof span → the three-method analysis → steel + links; wall load path; HW3 line-load check on the pressure slab

PART K — KNOWN ISSUES, ASSUMPTIONS AND UNCERTAINTIES

K.1 UNRESOLVED — do not guess

#	Item	Competing values	What closes it
U1	Sentry post base shear (C9)	STAAD 73.18 kN vs hand 59.3 kN (23 % apart, not reverse-engineerable: $73.18/624.5 \rightarrow A_h \approx 0.117$, not 0.100)	Print the STAAD seismic-weight summary. Design already uses the higher value.
U2	Is a direct hit a requirement?	Not stated anywhere	Military representative sign-off. Changes cover depth and burster design entirely.
U3	Design basis threat yield	50 psi / t_d 0.13–1.33 s stated, but the yield behind it is not	Client confirmation. Drives prompt-radiation cover depth.
U4	STAAD analysis/design command blocks	Never shown	Upload the .std files
U5	Underground model plate table and support definition	Never shown	Upload the .std files
U6	Node 213, sentry post	Row visible, values cut off	Upload the .std file
U7	Beam 16 (211→212), sentry post	Not visible; required for a closed frame	Upload the .std file
U8	Roof projection + parapet 4.162 kN/m	Given on the framing plan, not independently derived	Recompute from the parapet detail

K.2 ASSUMED — must be confirmed before construction

#	Assumption	Impact if wrong	Verify by
A1	Rockhead 1.5–2.0 m, competent below	Founding level, excavation cost	Site investigation
A2	GWT (–)2.000	Uplift, flotation, waterproofing class, wall design	Monsoon-season monitoring
A3	SBC 3240 kPa	Footing and mat sizing (both $\ll$ 13 % utilised)	Plate load / core testing
A4	k_s 100 000–500 000 kN/m³	Mat moments — sensitive. Run both bounds	Plate load test
A5	$K_o = 0.50$, γ 20/21	Wall lateral load	Site investigation
A6	$K_a = 1.0$ saturated (used); $K_a \approx 0.5$ dry berm (not relied on)	Headhouse wall load — removed from the critical path by C10	Would allow a reduction if measured
A7	Soak-pit absorption 20 L/m ² /day	Soak pit will not work if lower — likely on basalt	Percolation test, IS 2470 Pt 2 Cl. 4 — MANDATORY
A8	Structural seepage 0.5 L/m ² /day	Sump storage (currently 8.4 days)	Packer permeability tests
A9	Sentry infill NOT separated $\rightarrow R = 3.0$	$V_b \times 1.67$ if separated	Architect's decision
A10	$k_1 = 1.08$ (100-yr wind life)	Wind (does not govern)	Client brief
A11	$b_{eff} = 2.5$ m for the HW3 line load	HW3 strip check (26 % / 41 %)	Refined FE if ever critical
A12	Trapezoid factor 0.7946 used for both BM and FEM	~ 2 % on B2 support moment	Frame model
A13	Sentry post STAAD has no member releases	Frame moment distribution	Upload the .std file
A14	Poisson's ratio 0.20	Plate behaviour, minor	Standard

K.3 CONFIRMED but requiring attention

#	Item
F1	Protective boundary was not continuous — resolved by M1, awaiting client approval of the 400 mm box lengthening
F2	Stair-void cantilever $M(\text{root})$ 758.6 > midspan 700.2 — resolved, links added
ERR-1	My own yield-line coefficient error, 324.4 $\rightarrow$ 162.2 — corrected; no reinforcement change
Flotation	FoS 0.33 at the mat-only stage — a design output, mitigation is mandatory and on S-02
Deccan basalt resistivity 10^3 – 10^4 $\Omega \cdot \text{m}$	Test earth resistance early — affects EMP and lightning protection
Airlock purge 13 min	4–5 persons/hour — a manning constraint, must be on the drill card
EMP rebar cage 0 dB @ 1 GHz	Say it before a reviewer does. Zone 2 welded steel room is the answer

PART L — FINAL AUTHORITATIVE DESIGN REGISTER

Quick reference for the next session

REVISION	Architectural Rev F · Report Rev D · Structural Phase 2 Rev A + M1
BOX	22 000 × 6 200 external · 20 800 × 5 000 internal walls 600 · roof 900 · mat 600 · PCC 100 · clear height 3 200 cover 2 000 layered = 40.65 kPa
LEVELS	grade 0.000 · T/slab (-)2.000 · soffit (-)2.900 · floor (-)6.100 u/s mat (-)6.700 · formation (-)6.800 · GWT (-)2.000
INTERNAL WALLS	W5 200 @ 12600–12800 · W6 400 @ 14800–15200 · W7 400 @ 18000–18400
SHAFT / VOID / PAD	15200–18000 × 600–5600 / × 600–3760 / × 3760–5600
ESCAPE SHAFTS	ESC 1 (2050, 2050) · ESC 2 (19900, 2050) · 1400 dia, 250 collar
HEADHOUSE	14000–18000 × 600–5600 internal · walls 400 · roof 500 · top +0.900
ENTRY STAIRWELL	9500–15800 × 6000–7500 internal · 12R @ 166.667/300 · waist 250
SENTRY POST	4000 × 5000 · grid 3650 × 4650 · C1 350 ² · B1/B2 250 × 450 · S1 150 F1 1500 ² × 600 · levels +0.450 / +3.650 / +6.700 / +7.000
MATERIALS	Box M35 / Fe500D · Sentry M30 / Fe500 Ec 29 580 / 27 386 N/mm ² · cover 75/50/40 (box), 30/40 (sentry) Ld 40φ (M35) / 46φ (M30) · lap 50φ staggered
SOIL	Basalt · SBC 3240 kPa [A] · K0 0.50 · γ 20/21 · GWT (-)2.000 [A] ks 100 000–500 000 kN/m ³ [A] — RUN BOTH BOUNDS
BLAST	p_so 344.7 kPa (50 psi) · td 0.13–1.33 s · μ 5 · DLF 1.111 DESIGN 383 kPa on roof AND walls (Ka 1.0) fck,dyn 43.75 · fy,dyn 625 · NO dynamic increase on shear T 13.4 ms · td/T = 10–100 → QUASI-STATIC
ROOF TOTAL	448.15 kPa (383 + 40.65 + 2.0 + 22.5), no live load (Cl. 11.2)
HEADHOUSE ROOF	396.5 kPa · HEADHOUSE WALLS 383 kPa either face (C10)
SEISMIC	Box Ah 0.075, Vb 1050 kN (negligible) Sentry Ah 0.100, R 3.0, **Vb 73.18 kN (STAAD)** vs wind 29.9 kN
COMBINATIONS	Box 101/102/**103 BLAST**/104/105 · Sentry 101–113, 201, 202
REINFORCEMENT — THE SIX THAT MATTER	
ROOF	T25 @ 150 EF EW · T12 4L @ 250 end 1500 / 2L @ 300 mid 51 %
WALLS	T16 @ 150 EF EW · T12 closed @ 200 68 %
W6/W7	T20 @ 150 EF EW · T12 4L @ 200 69 %
MAT	T16 @ 150 EF EW · T12 @ 250 × 250 grid 84 %
HH ROOF	T20 @ 150 EF EW · T12 4L @ 175 / 250 90 %
HH WALLS	T16 @ 150 EF EW · T12 4L @ 250 59 %
SENTRY	S1 T8 @ 150 B/W + T8 @ 300 edge top + T8 @ 200 torsion 700 ² × 4 corners B1 4-T16 / 2-T16 · B2 3-T20 / 2-T20 · T8 @ 100 hinge / @ 150 C1 8-T16 · T10 hoops + cross-ties @ 85 over 500 · biaxial 0.819 F1 T12 @ 150 B/W
STATUS	STAAD 2 models exist; .std NOT uploaded; screenshots only DXF 8 output sheets S-01..S-08, validated, + A1 PDFs CALCS Complete to IS 456; SDOF support rotation NOT done (Phase 3) DRAWINGS Sentry beams/columns/footings NOT yet drawn (would be S-09)

PART M — INSTRUCTIONS FOR CONTINUING THIS PROJECT IN A NEW CLAUDE SESSION

- 1. Read this entire MASTER PROJECT STATE before touching anything.** Do not skim to the section that looks relevant. The conflicts and the dependency map matter more than any single value.
- 2. Treat PART A (Current Authoritative Project State), PART B (Structural Design) and PART L (Final Register) as the primary source of truth.** Everything else supports them.
- 3. Treat superseded revisions as historical only.** Part H tells you what was superseded and why. **Critically: every uploaded input DXF is at the PRE-M1 geometry (box 21 600). The M1 geometry exists only in the output drawings and in this document.** Do not read a current dimension off an input DXF east of X = 14 800.
- 4. When project files are uploaded, compare them against this document before using them.** Parse them; do not assume.
- 5. If a file conflicts with this document, STOP and identify the conflict.** Present both values, say which this document holds, and ask. Do not silently adopt either.
- 6. Never guess missing engineering information.** If something is marked **[NOT AVAILABLE]** or **[UNRESOLVED]**, it stays that way until the user supplies it. The correct response is to ask, not to fill the gap.
- 7. Preserve the existing design unless a change is explicitly requested.** This design is the product of ten resolved conflicts, two structural findings and one corrected arithmetic error. Do not “improve” it casually.
- 8. When changing one parameter, work the dependency map in Part J** and list every downstream item that must be updated. State that list before making the change.
- 9. Maintain consistency across geometry, calculations, STAAD, DXF, reinforcement and drawings.** The rule this project has held throughout: **no drawing shows an arrangement that differs from a calculation.** If you change steel, change the drawing in the same turn.
- 10. Give every significant change a revision identifier** (M2, M3...; C11, C12...; ERR-2...) and add it to the tables in Part H.
- 11. Never overwrite a previous revision without preserving it.** Add rows to Part H; do not edit them away.
- 12. To regenerate a drawing, edit the Python generator and re-run it — do not edit the DXF.** See Part E.4. proj.py holds every geometric constant; changing a dimension there propagates to all eight sheets. Always re-run validate.py and render.py afterwards and **look at**

the render before delivering.

13. When generating any new STAAD content, keep it consistent with Part A geometry and Part D conventions — remembering the underground model's **6.700 m Y-offset** and its **mid-surface** geometry.

14. Before finalising any revision, cross-check in this order: geometry → materials → loads → structural model → analysis → design calculations → reinforcement → drawings → revision number. A change that stops at “reinforcement” and never reaches “drawings” is not finished.

15. Label everything you produce as newly calculated, reconstructed, assumed or unresolved, using the same tags as this document.

16. Do not claim a design is verified merely because an earlier version existed. Recompute. The C2 headhouse roof — a calculation carried forward for two revisions while the geometry changed underneath it — is exactly why.

17. Preserve the design intent. In particular: - The **protective boundary is Blast Doors 1 and 2 at (–)6.100**. Everything above them is expendable, and that is deliberate. - **The roof spans one-way across the 5 000 mm width**. Lengthening the box is free; widening it is expensive. This single fact underpins Modification M1. - **No movement joints anywhere inside the protective envelope**. A movement joint is a guaranteed blast, gas and EMP discontinuity. - **IS 4991 Cl. 10.3.1.1 — no dynamic increase on shear**. This is the most commonly mis-applied rule in blast design and it is why several elements have links they would not otherwise need. - **Say the weakness before the reviewer finds it**. That principle produced F1, F2, C9, C10 and ERR-1, and it is the reason this package is defensible.

18. Two things that must never be claimed: - that a static STAAD run proves blast resistance (it gives the *demand* — the support-rotation check is Phase 3); and - that flotation is satisfied because the model shows no uplift (the springs take tension — it is a hand check).

END OF MASTER PROJECT STATE FILE. Compiled 2 September 2026 from the complete Phase 1 and Phase 2 record. Every value is tagged CONFIRMED, RECONSTRUCTED, ASSUMED, UNRESOLVED or NOT AVAILABLE. Nothing in this document was invented.